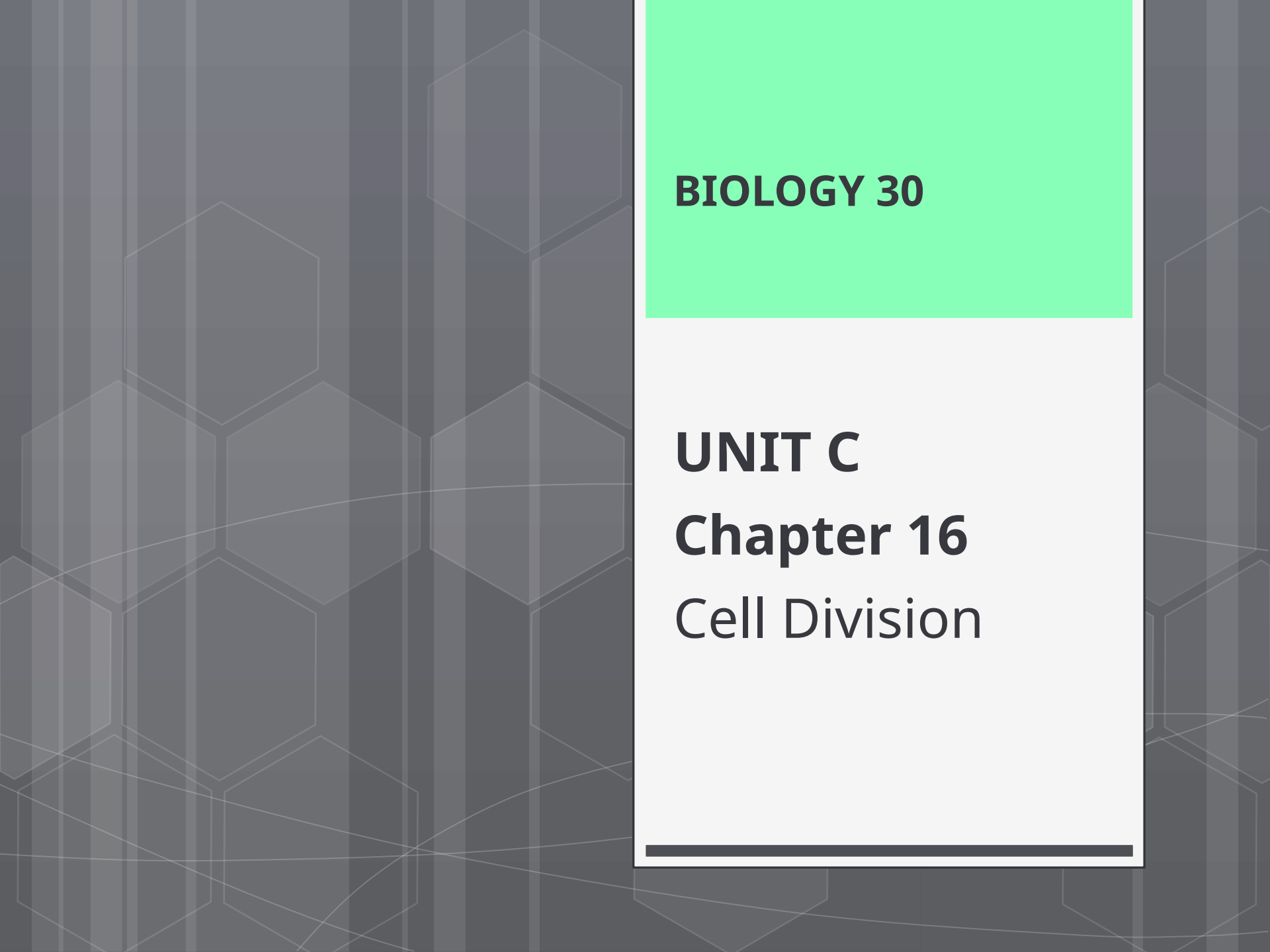




BIOLOGY 30

UNIT C

Chapter 16

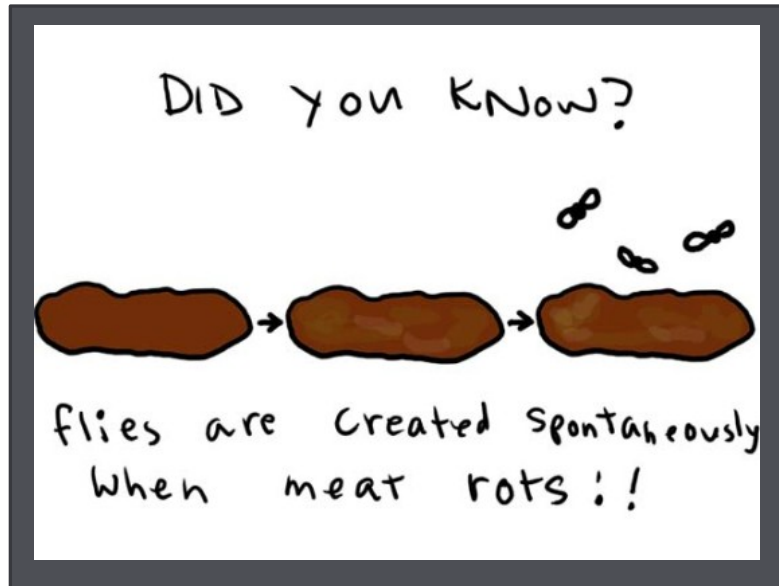
Cell Division

TOPIC 1: MITOSIS

- somatic cells vs. sex cells
- autosomes vs. sex chromosomes
- chromosomes vs. chromatin
- homologous chromosomes vs. sister chromatids
- haploid (n) vs. diploid ($2n$)
- stages of interphase: G1, S, G2
- stages of mitosis: prophase, metaphase, anaphase, telophase
- cancer, cloning, and stem cells
- telomeres & aging

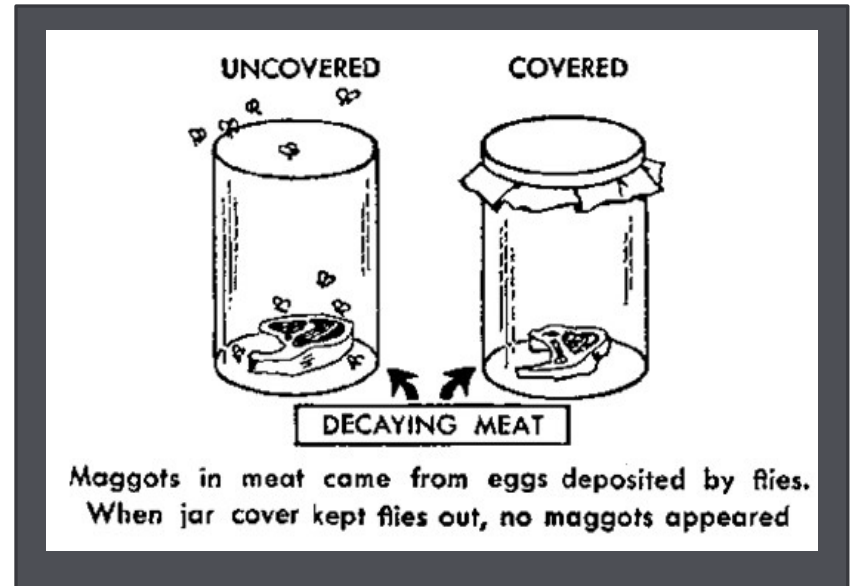
Textbook pgs. 548-562

What we used to think:



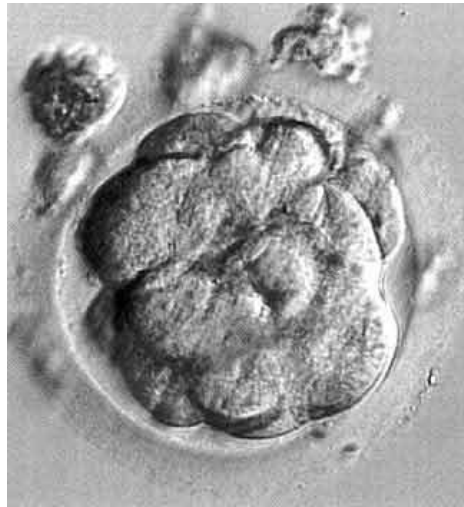
- Life can arise from non-living things (abiogenesis)

What we now know:



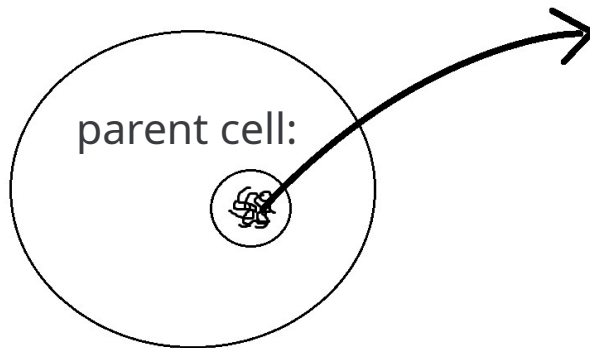
- Life arises only from the division of living cells (biogenesis)

From the moment of fertilization, cells are continuously growing, dying, and being replaced...



... this is achieved through the process of cell division. But before we get into too many details, we have some background info to talk about..

Each new cell, or ***daughter cell***, must contain the same genetic information as the original cell, or ***parent cell***

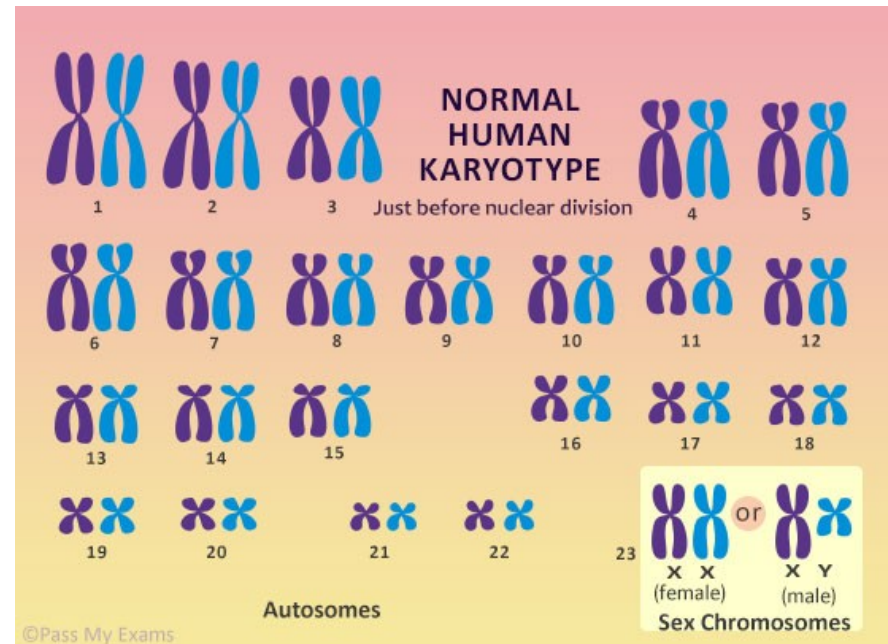


DNA is this genetic information. It is found within the nucleus of each and every cell in the body.

Creating identical daughter cells is accomplished through the processes of DNA replication and cell division.

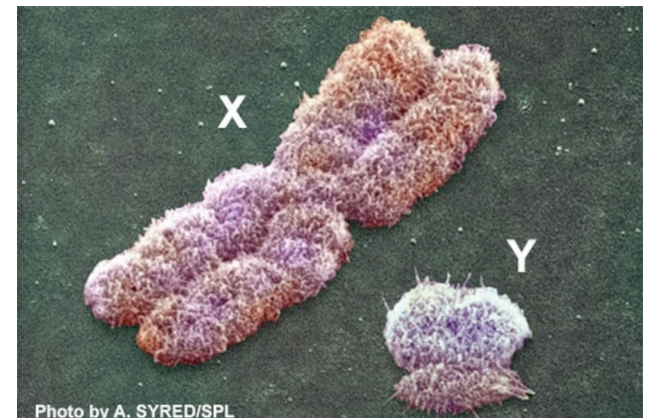
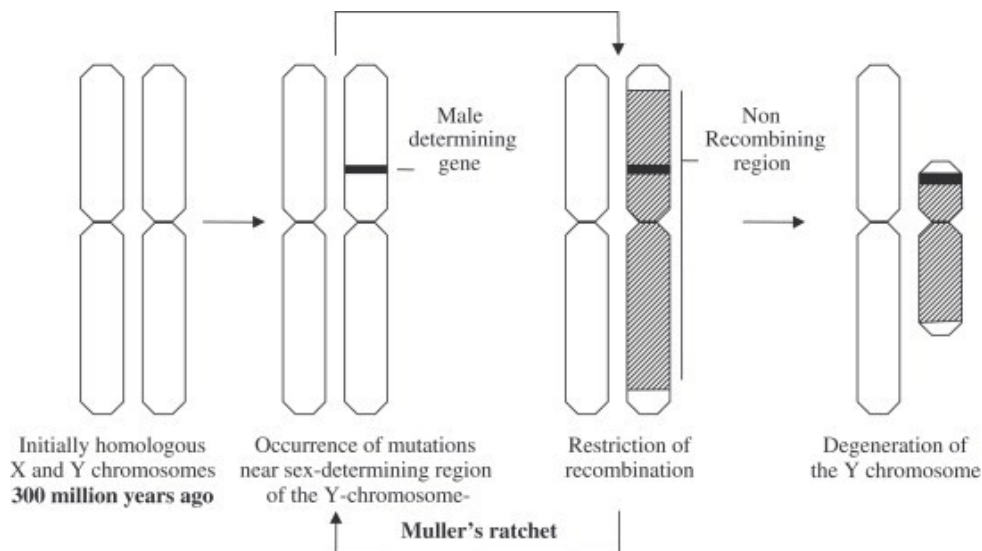
- The best way to visualize chromosomes is a **karyotype**, which is an image of all the chromosomes the nucleus.
- The **somatic cells** (body cells, ie. not gametes) of humans have 23 pairs of chromosomes or 46 chromosomes in total.
 - In females, there are 23 pairs of **homologous chromosomes** (22 pairs of **autosomes** and 1 pair of X **sex chromosomes**)
 - In males, there are 22 pairs are **homologous chromosomes** (**autosomes**), and 1 pair of **non-homologous sex chromosomes** (“**sex chromosomes**”)

- **Homologous chromosomes** are chromosomes with the same gene sequence as one another; one homologue is of maternal origin, and the other is of paternal origin.
- Although homologues carry the same set of genes, they may carry different forms of the each gene (a.k.a **alleles**)
- Because human cells carry two pairs of each chromosome, we are said to be **diploid**

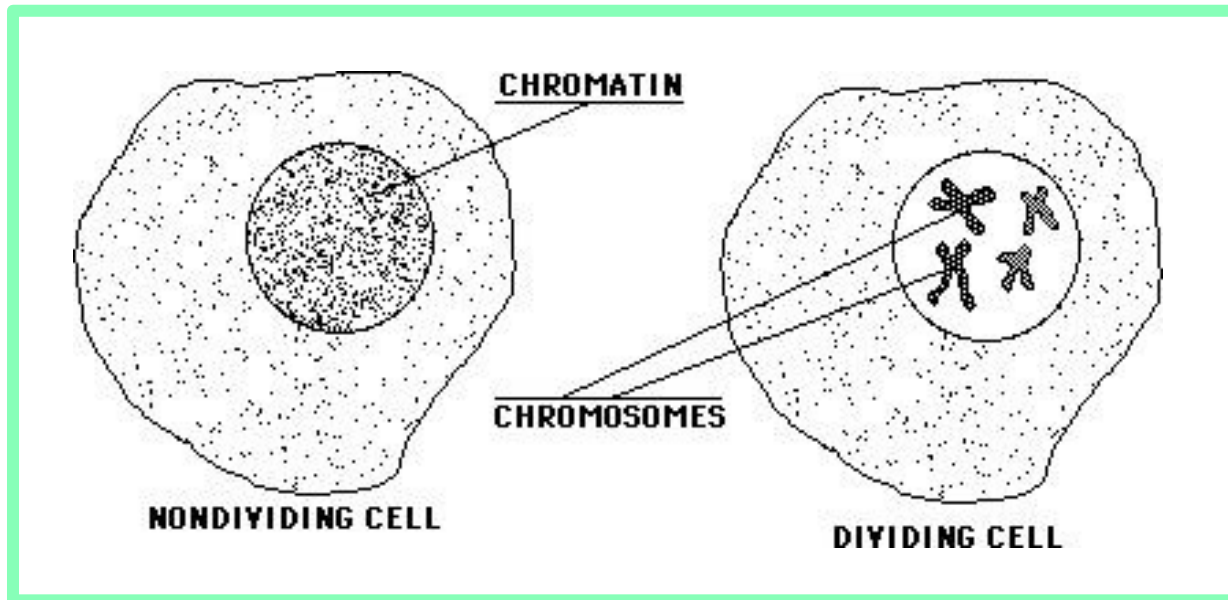


Why is the Y-chromosome smaller?

The **Y chromosome** is smaller because it has lost many genes over time. It started as a pair with the X chromosome, but over millions of years, it stopped recombining with the X and many of its genes degraded. Now, it mainly carries a few genes essential for male traits, like the **SRY** gene for sex determination.

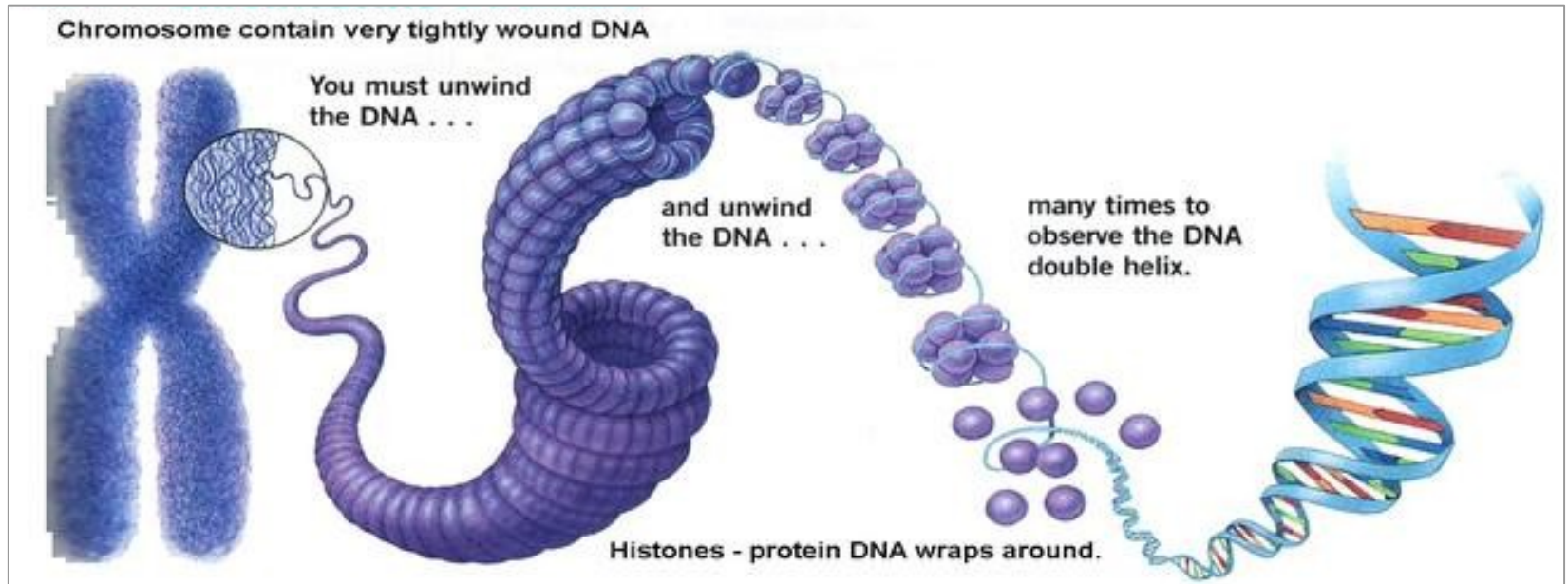


“Unpackaged” DNA is referred to as
chromatin:

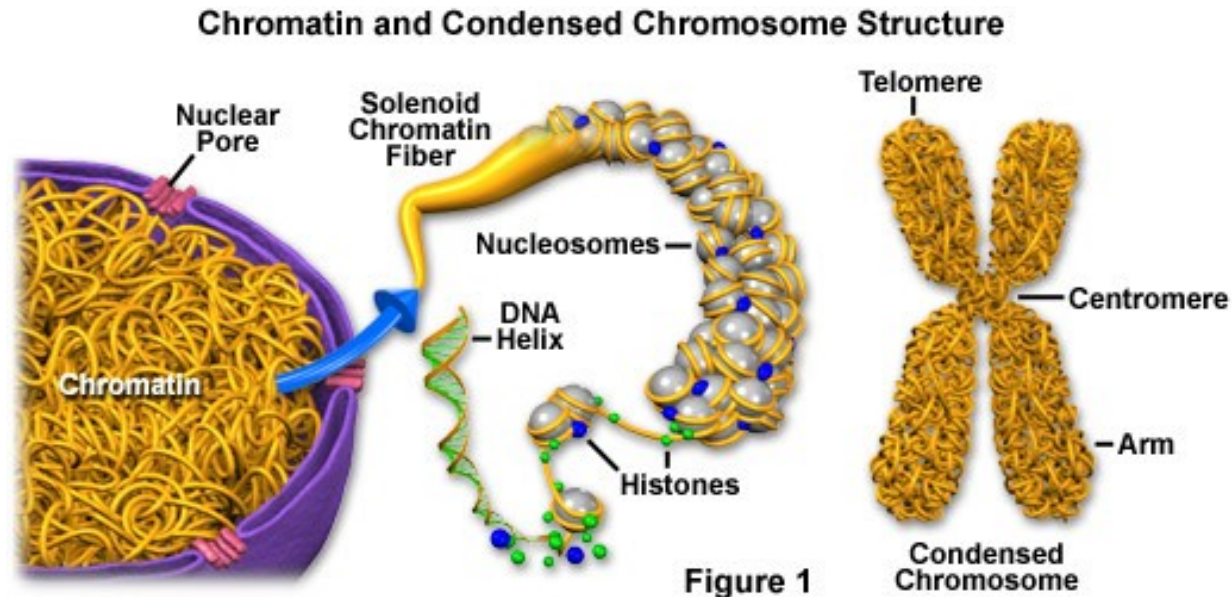


DNA exists as chromatin during most of the cell's life but **condenses into chromosomes** when the cell is about to divide.

Chromosomes contain very tightly wound DNA.



Before cell division, it is important for chromatin to condense into chromosomes in order to be able to separate them without damage and breakage. Image how hard it would be to separate cooked spaghetti that is all entangled!



Chromosome Number

varies from species to species...



Humans = 46

Bananas = 22



Dogs = 78



Ferns = 1260

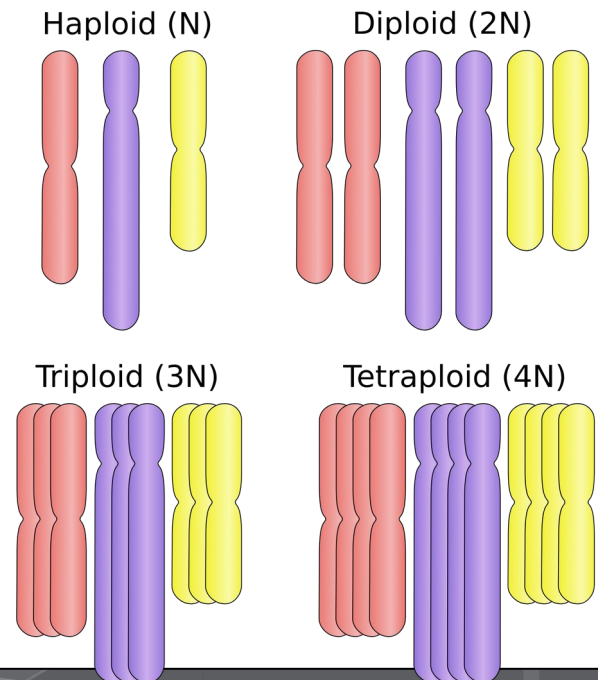
More chromosomes DOES NOT necessarily result in
higher intelligence or complexity

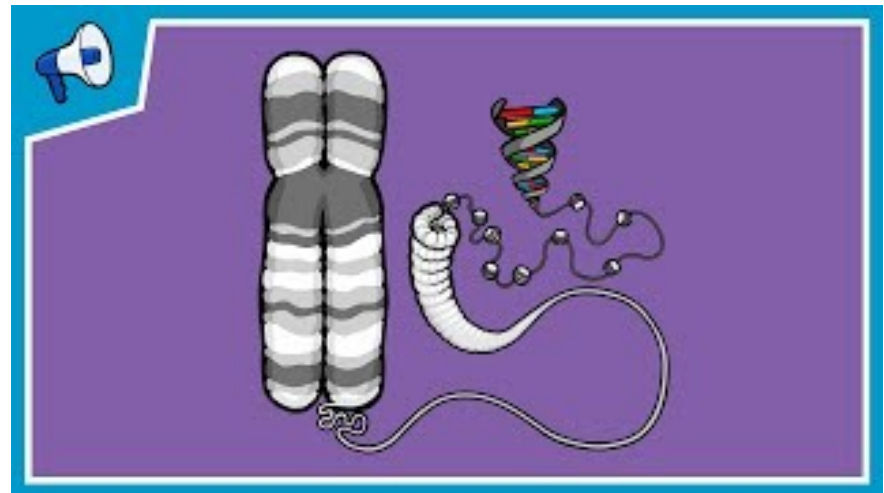
Ploidy

- **Ploidy** refers to the number of sets of chromosomes in a cell.
- Human somatic (body) cells are **diploid, or “2n”**
 - This means there are two copies of each chromosome (one of maternal origin and one of paternal origin)
- Human gametes are **haploid, or “n”**

$$2n = 46$$
$$n = 23$$

Some organisms, such as earthworms, contain more than two homologous chromosomes in each cell. These organisms are said to be ***polyploid***.







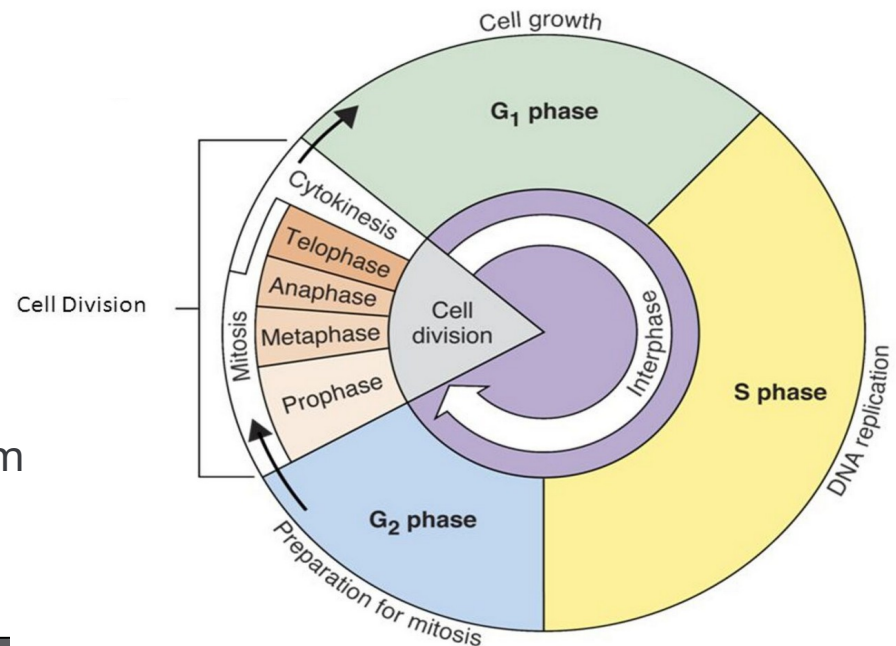
Alright..now that we have some background understanding of concepts relevant to the cell cycle and division, let's get into it!

The Cell Cycle Overview

Almost every cell goes through distinct phases and subphases

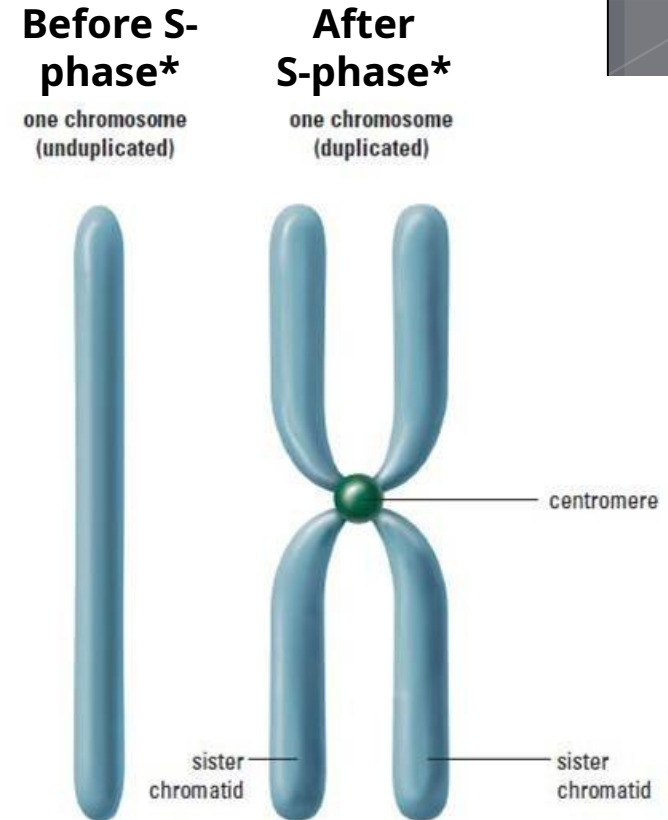
1. **Interphase** → makes up most of the cell cycle, the cell is not dividing
 - a. **G₁ Phase** (cell growth and normal metabolic processes)
 - b. **S Phase** ('S' for synthesis → DNA replication takes place)
 - c. **G₂ Phase** (preparation for division)

2. **M phase (cell division)**
 - a. **Mitosis** (nuclear division)
 - i. **Prophase**
 - ii. **Metaphase**
 - iii. **Anaphase**
 - iv. **Telophase**
 - b. **Cytokinesis** (division of cytoplasm and organelles)



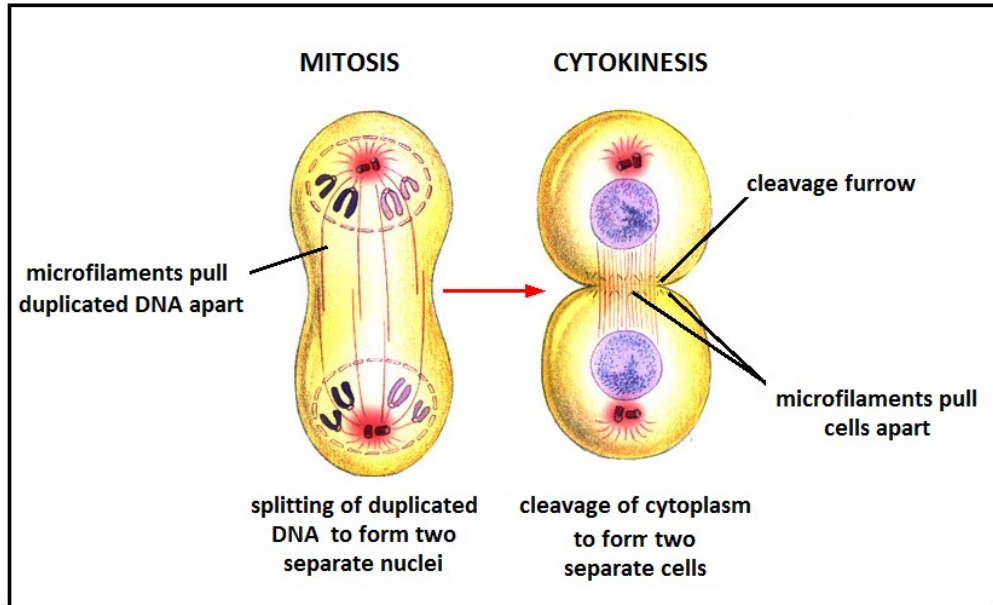
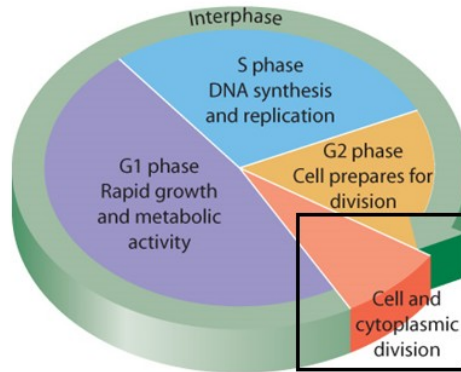
Duplicated Chromosomes

- During the S-phase of interphase, DNA is replicated to double the genetic content of the cell. This is to ensure each daughter cell received a copy of each chromosome and the genes on it.
- We can observe this best when looking at **unduplicated** and **duplicated chromosomes**.
 - The two arms of a duplicated chromosome are called **sister chromatids** and are exact copies of one another (result of S-phase)
 - Sister chromatids are held together by the **centromere**



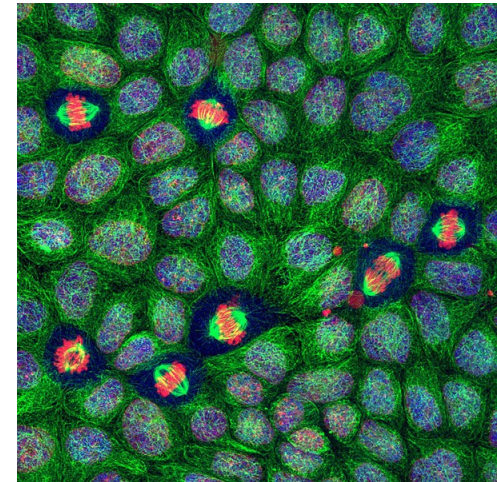
**Note: Remember that chromosomes are not actually condensed like this until cell division is about to occur, this is just for illustration purposes.*

The Cell Cycle

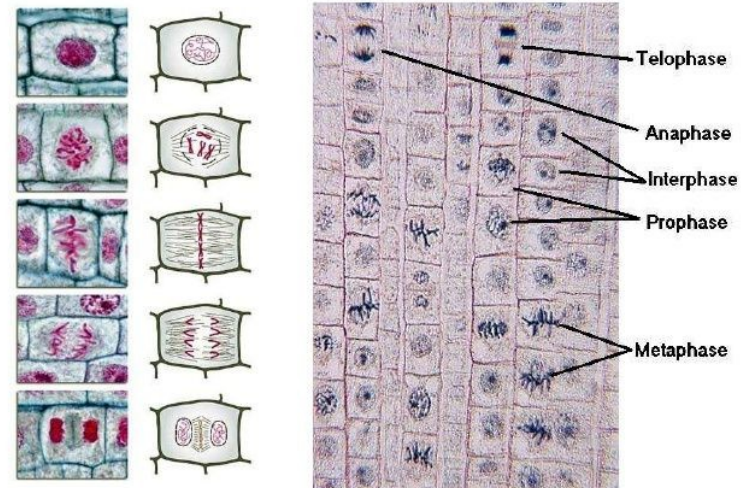


M-phase accounts for the actual division of genetic information (*mitosis*) and splitting of the cytoplasm itself (*cytokinesis*)

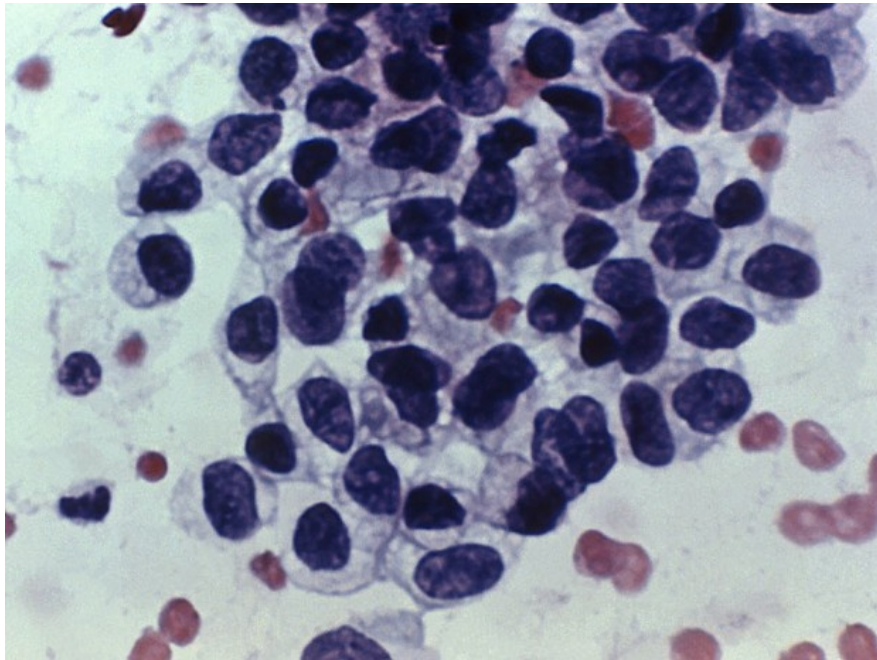
- If you were to look at a sample of healthy tissues under a microscope, most of the cells you would observe would likely be in interphase/early prophase. Why do you think that is?
- If cells spend most of their time (approx 90%) in interphase, the chance of finding a particular cell in interphase is much greater. Therefore most cells should be found in interphase at any time of observation.
- Actively growing tissues, such as the the tip of plant roots, however will have a slightly higher proportion of cells in mitosis



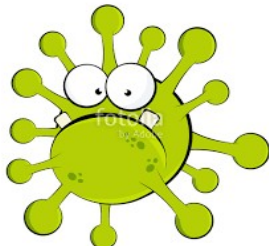
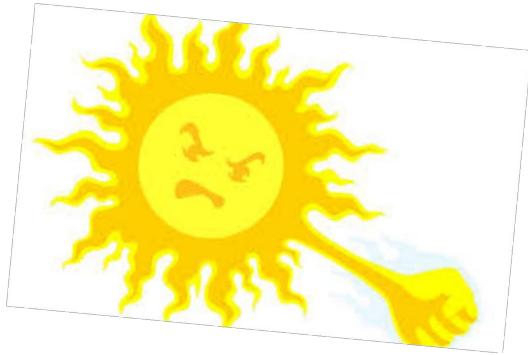
Mitosis in Onion Root Tips



What would happen if cells were constantly dividing in an uncontrolled fashion?



CANCER



Cancer & the Cell Cycle

- Uncontrolled cell division may be caused by exposure to ***mutagens***, or agents which change an organism's genetic material.
- Mutagens may be physical, chemical, or biological in origin.
- Mutagens that specifically lead to the formation of cancerous cells are further classified as ***carcinogens***.

Mitosis

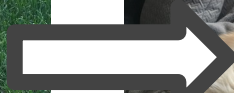
Is a form of cell division that produces daughter cells that are identical to parent cells.

PURPOSES:

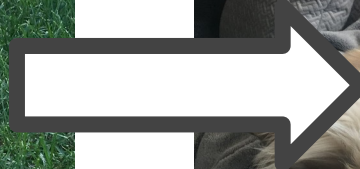
- Growth
- Maintenance
- Repair



The extent of repair varies from organism to organism

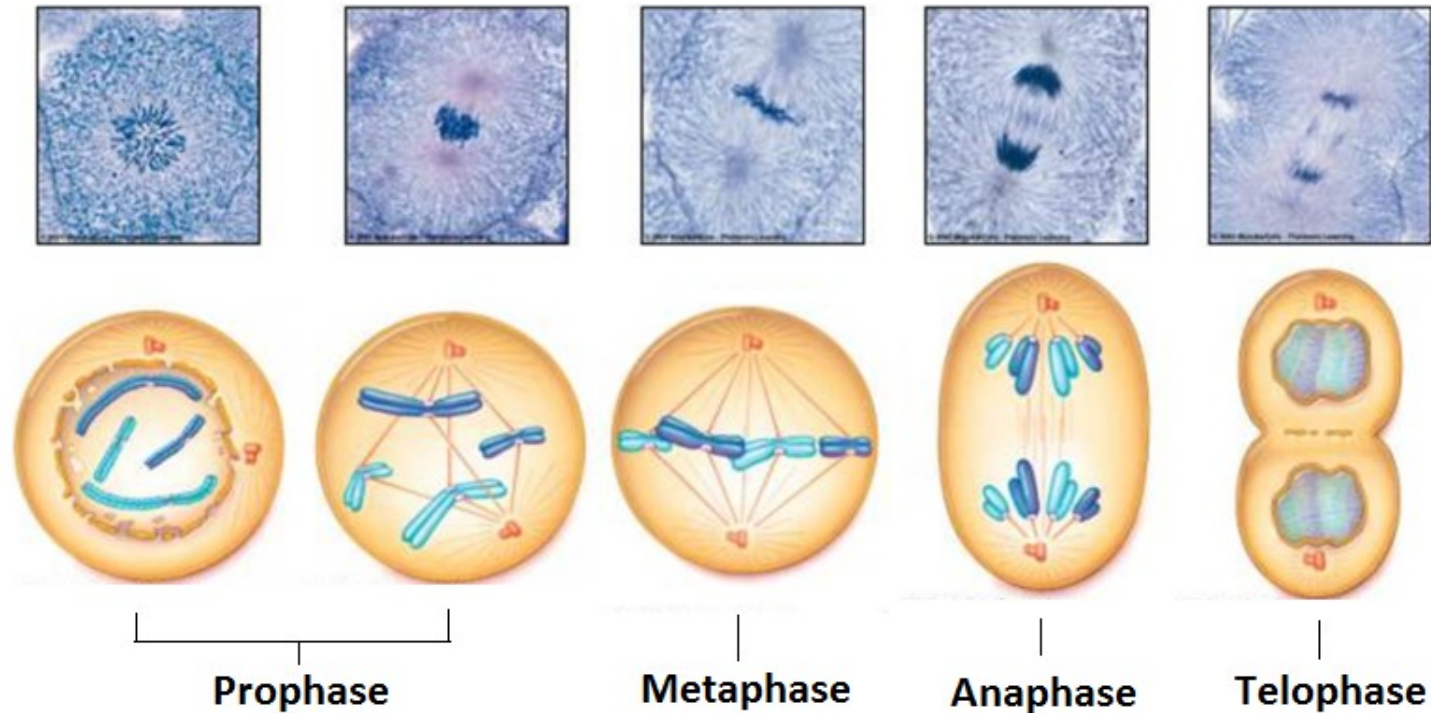


Mitosis & Growth



Mitosis

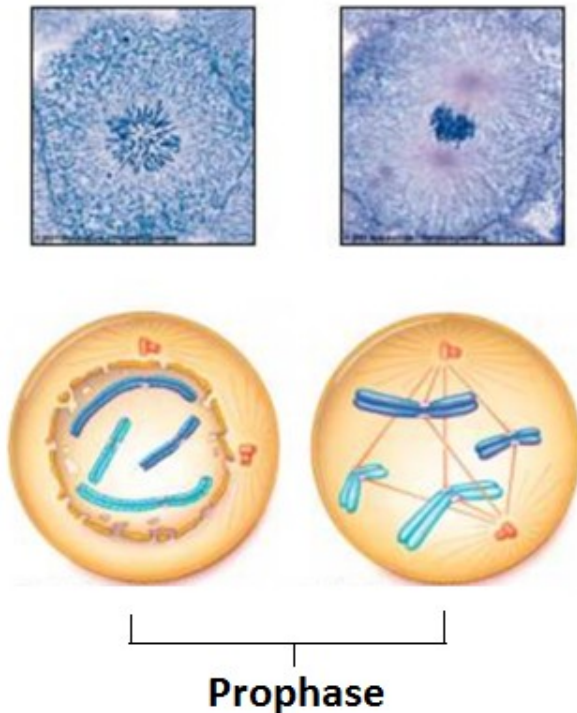
After DNA replication (interphase), cells go through four stages of mitosis before cytokinesis can occur:



PMAT! (Your dog peed on the mat..)

1. PROPHASE

At this stage, each pair of homologous chromosomes has already been duplicated, so the cell contains twice as much DNA than normal.

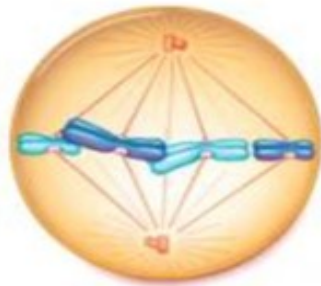
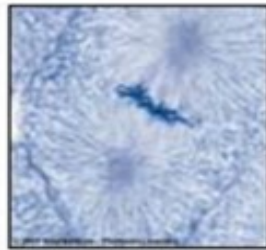


Key events:

- Chromatin condenses to form chromosomes
- Nuclear membrane breaks down
- Centrioles move to opposite poles of the cell and spindle fibres begin to form (spindle fibres are required to eventually pull the duplicated chromosomes apart)

NOTE: These examples only show 4 chromosomes ($2n=4$), but remember that a actual human cell would be showing 46 chromosomes ($2n=46$)

2. METAPHASE



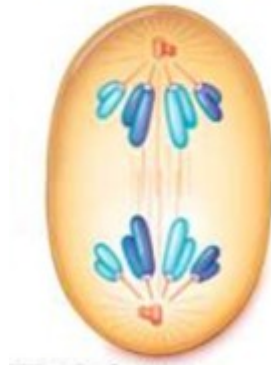
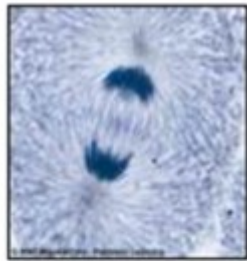
Metaphase

Key events:

- Spindle fibres attach to the centromere of each chromosome, guiding them to the equator (middle line) of the cell
- Duplicated chromosomes line up in single file, forming the “metaphase plate”

M for middle!

3. ANAPHASE

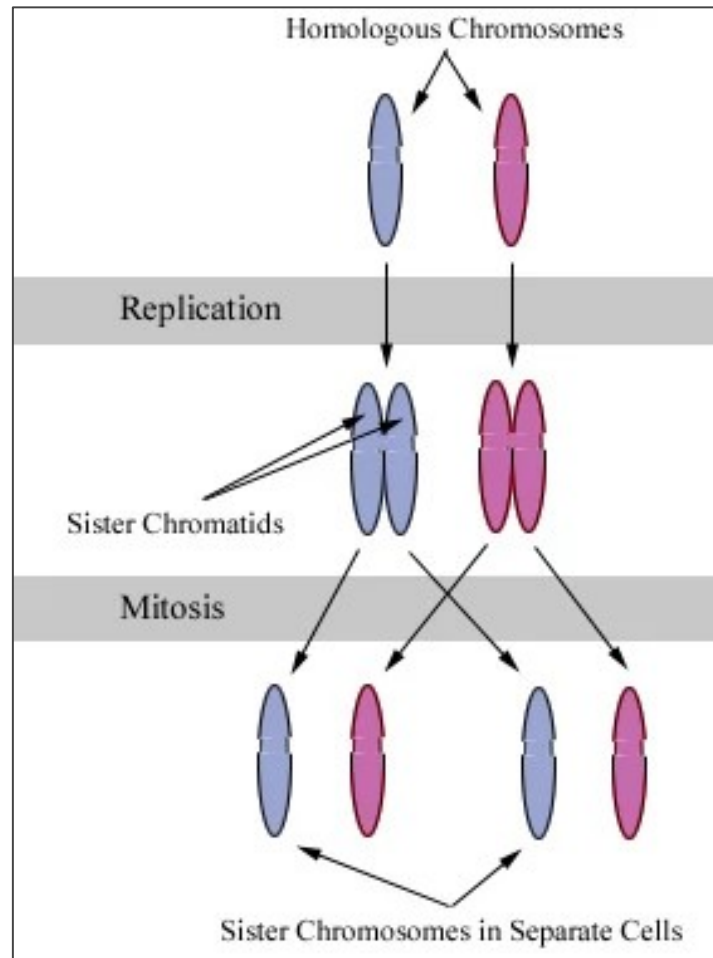


Anaphase

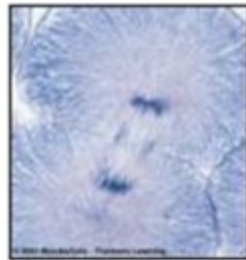
Key events:

- Spindle fibres shorten and sister chromatids are pulled apart from one another; centromeres split
- Two opposing poles are formed, each with a complete diploid set of DNA

3. ANAPHASE



4. TELOPHASE



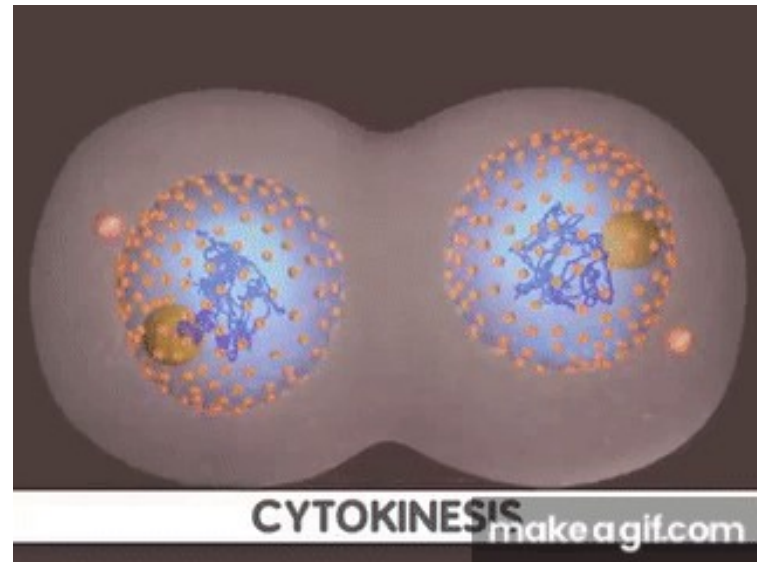
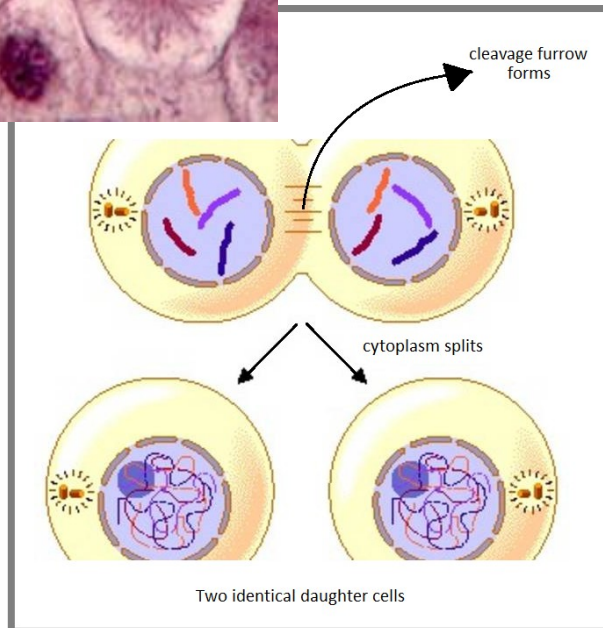
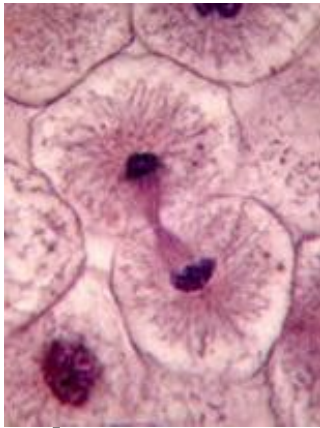
Telophase

Key events:

- Chromosomes reach opposite poles of the cell
- Spindle fibres break down
- Chromosomes begin to unwind into chromatin
- Nuclear membrane forms around each new set of genetic info.

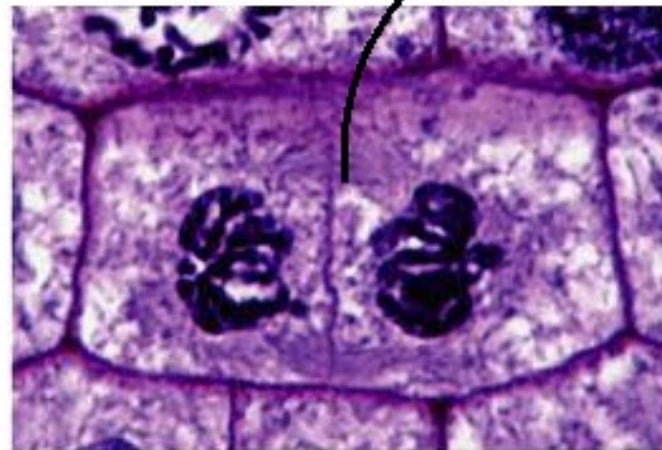
Cytokinesis

- o Cytoplasm, containing organelles, splits to form two identical daughter cells with the same genetic content as the original parent cell
- o Daughter cells are now in G1 of interphase



Telophase and cytokinesis in plant cells looks a bit different...

- o Because plant cells contain a rigid cell wall, cytoplasm will not furrow and pinch off
- o Instead, a cell plate forms between the two daughter nuclei, eventually developing into a cell wall



cell wall begins
to form

Mitosis in Plant Cells:



1 Prophase. The chromatin is condensing. The nucleolus is beginning to disappear. Although not yet visible in the micrograph, the mitotic spindle is starting to form.

2 Prometaphase. We now see discrete chromosomes; each consists of two identical sister chromatids. Later in prometaphase, the nuclear envelope will fragment.

3 Metaphase. The spindle is complete, and the chromosomes, attached to microtubules at their kinetochores, are all at the metaphase plate.

4 Anaphase. The chromatids of each chromosome have separated, and the daughter chromosomes are moving to the ends of the cell as their kinetochore microtubules shorten.

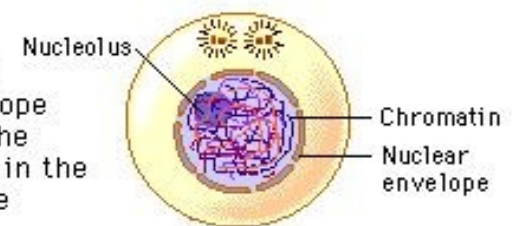
5 Telophase. Daughter nuclei are forming. Meanwhile, cytokinesis has started: The cell plate, which will divide the cytoplasm in two, is growing toward the perimeter of the parent cell.

Mitosis Recap



Interphase

The nucleolus and the nuclear envelope are distinct and the chromosomes are in the form of threadlike chromatin.



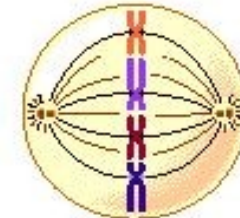
Prophase

The chromosomes appear condensed, and the nuclear envelope is not apparent.



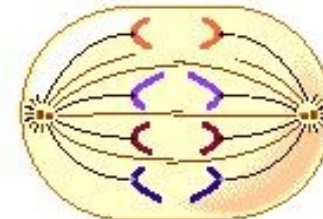
Metaphase

Thick, coiled chromosomes, each with two chromatids, are lined up on the metaphase plate.



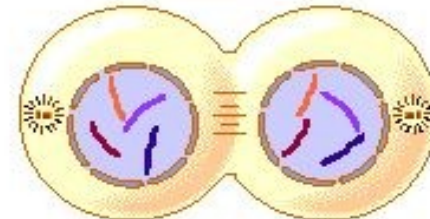
Anaphase

The chromatids of each chromosome have separated and are moving toward the poles.



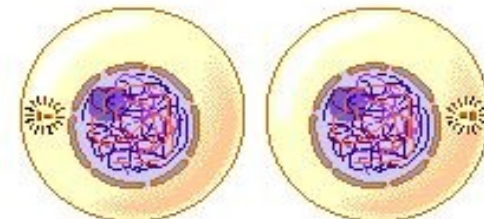
Telophase

The chromosomes are at the poles, and are becoming more diffuse. The nuclear envelope is reforming. The cytoplasm may be dividing.



Cytokinesis

Division into two daughter cells is completed.



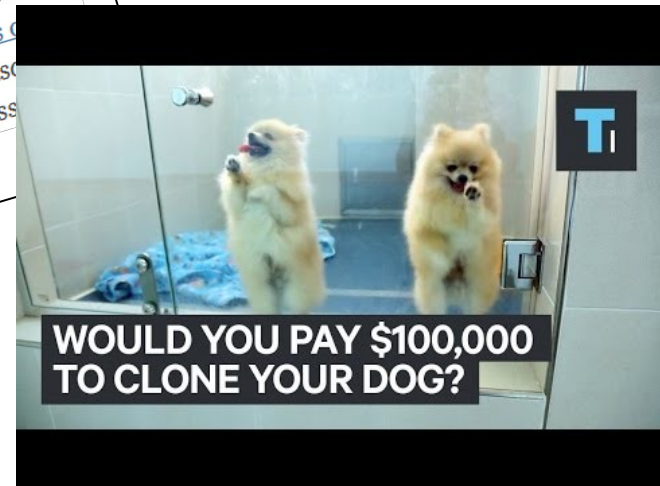
Extra:

The New York Times

Barbra Streisand Explains: Why I Cloned My Dog

March 2, 2018

In a frank and lengthy [interview in Variety](#) this week, Barbra Streisand dropped one very notable aside: that [two of her dogs were clones of her previous dog](#), Samantha, who had recently died. Here, Ms. Streisand explains how this medical marvel, born of sadness, came to pass.



Extra: Stem Cell Research

Most cloning-based research has focused on cloning individual cells, tissues and organs from ***stem cells*** (cells that have not yet become specialized)...

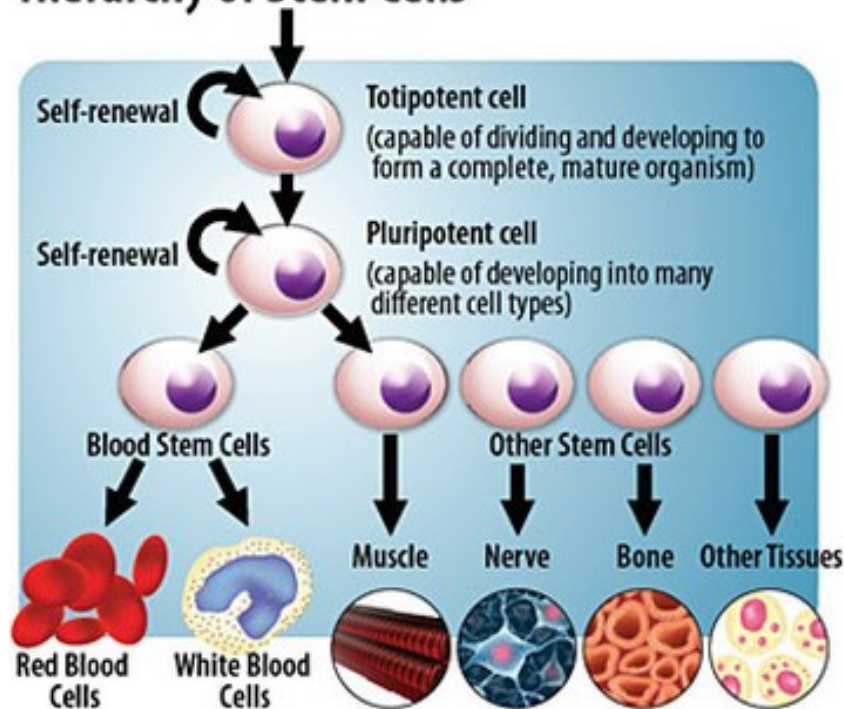


What might be some useful applications of this research???

Extra: Stem Cell Ethics

(More on this in Ch. 18)

Hierarchy of Stem Cells

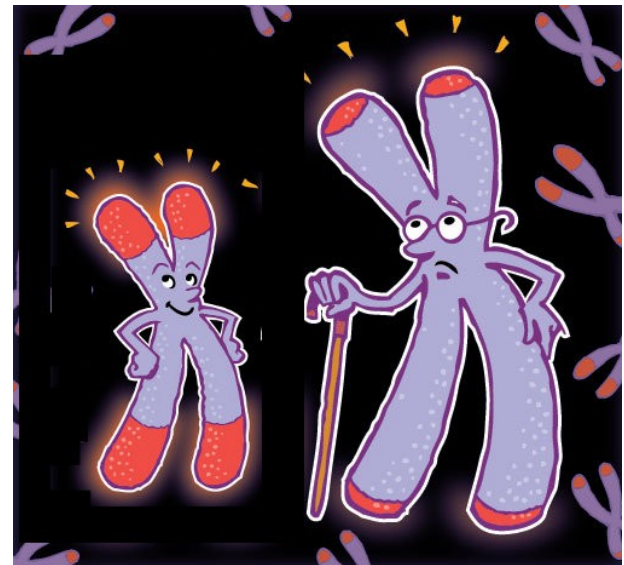


- Only a zygote is truly totipotent; it is considered unethical to obtain stem cells from this source.
- Pluripotent stem cells, however, can easily be obtained from adult tissues.

<https://www.genome.gov/25020028/cloning-fact-sheet/>

Extra: Mitosis and Aging

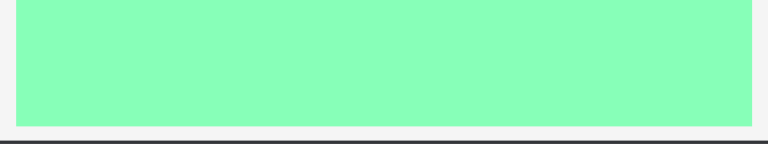
- Telomeres are caps found at the end of each chromosome. Every time a cell divides, its telomeres shorten.
- Eventually, the telomeres become so short that a cell will no longer be able to undergo mitosis, and cell death occurs.
- Telomere length thus serves as a sort of “molecular clock” for cellular aging.
- Lifestyle factors such as smoking, lack of physical activity, obesity, stress, exposure to pollution, etc. can potentially increase the rate of telomere shortening, cancer risk, and pace of aging.



Extra: Long-Term Consequences?



<https://www.newscientist.com/article/dn3393-dolly-the-sheep-dies-young/>



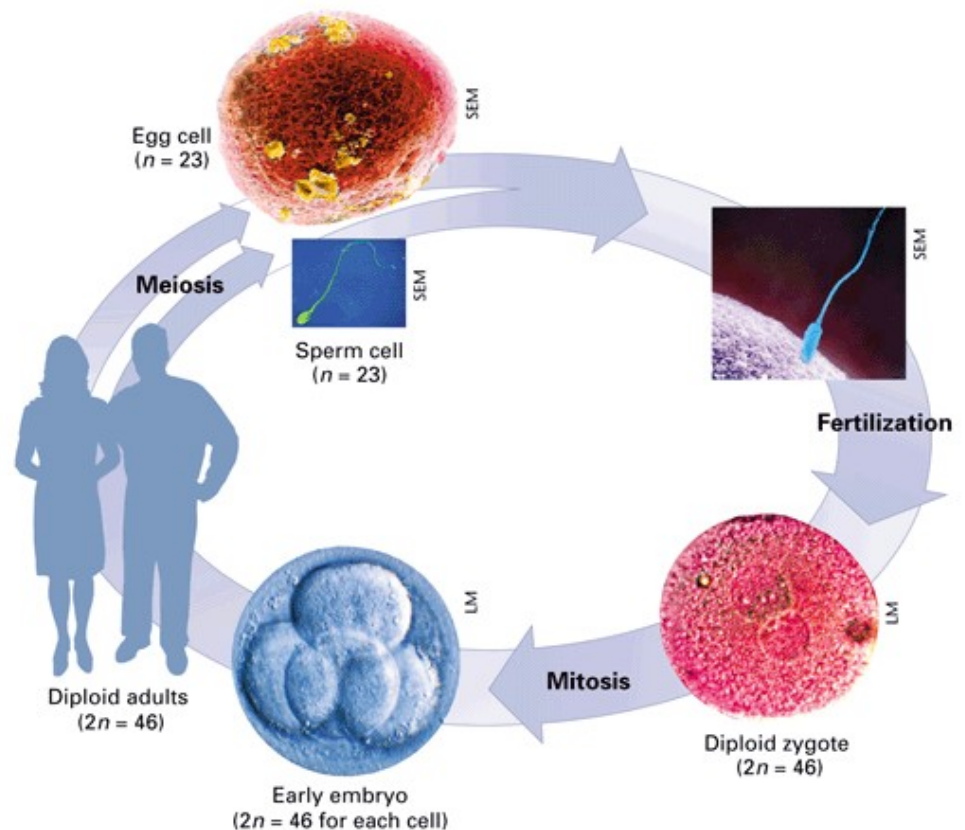
TOPIC 2: MEIOSIS

- Reduction of chromosome number
- Stages of meiosis
- Crossing over & independent assortment
- Mitosis vs. meiosis: similarities & differences
- Nondisjunction

Textbook pgs. 563-572

Meiosis

- While somatic (body) cells must contain a complete set of chromosomes, gametes possess just half the genetic information of parent cells.
- This is accomplished through the process of ***meiosis***.
- In humans, haploid gametes join during fertilization to form a diploid zygote.

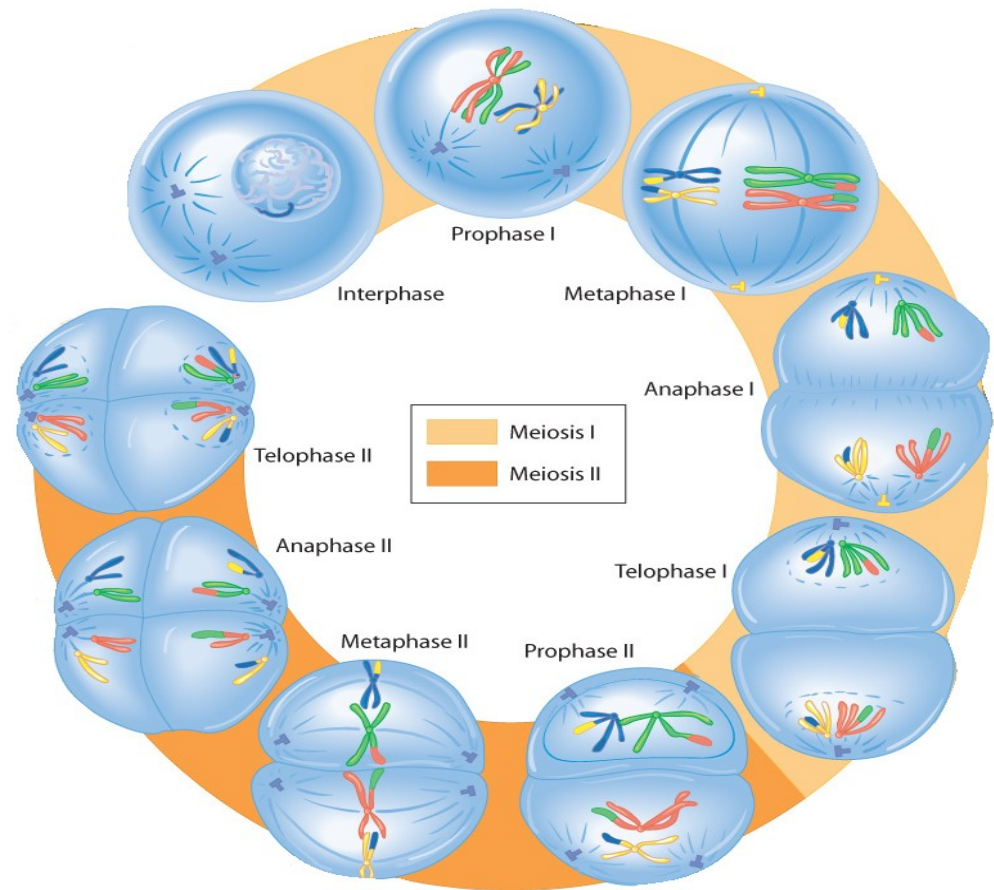
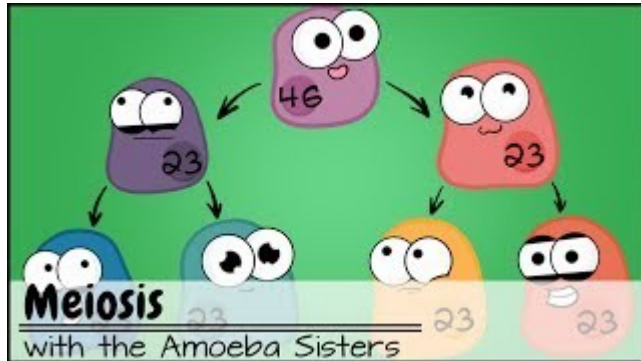


Meiosis

The outcome of meiosis is different than that of mitosis in three ways:

- 1) Produces **four non-identical daughter cells rather than two identical ones**
- 2) Daughter cells have **half the chromosomes** of parent cells
- 3) Daughter cells have **chromosomes with different combinations of genes** compared to those in parent cells
 - This is achieved through crossing over (more on this later)

Meiosis



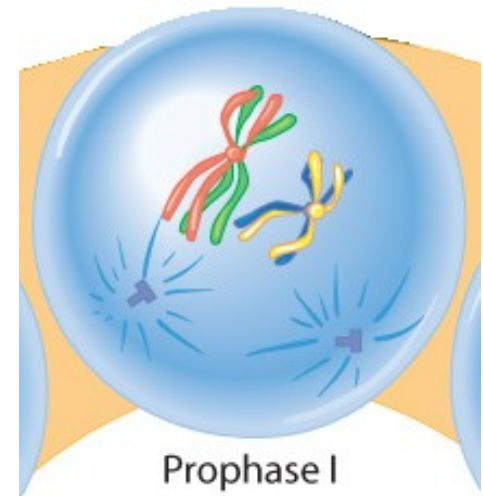
Involves the same stages of cell division as mitosis; however, in meiosis, **each stage occurs twice.**

PROPHASE I

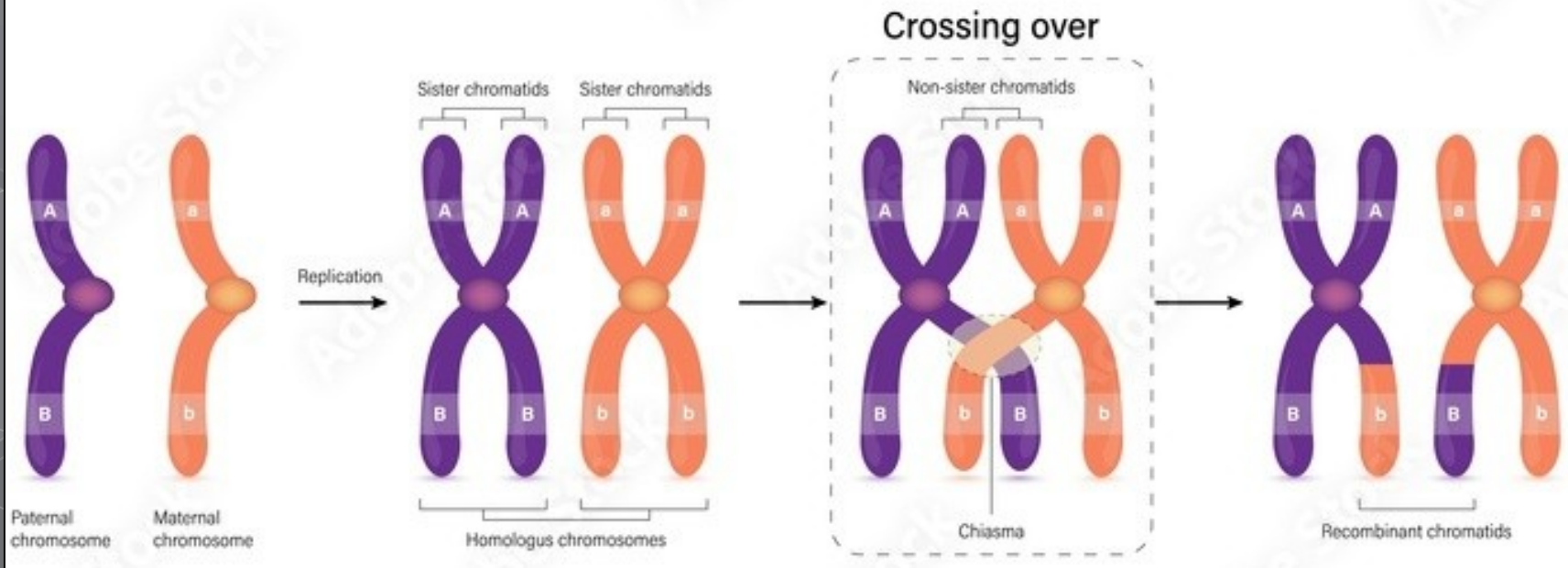
- Like in mitosis, meiosis begins with interphase, or DNA replication. Entering prophase I, then, the cell contains two pairs of homologous chromosomes.

Key events:

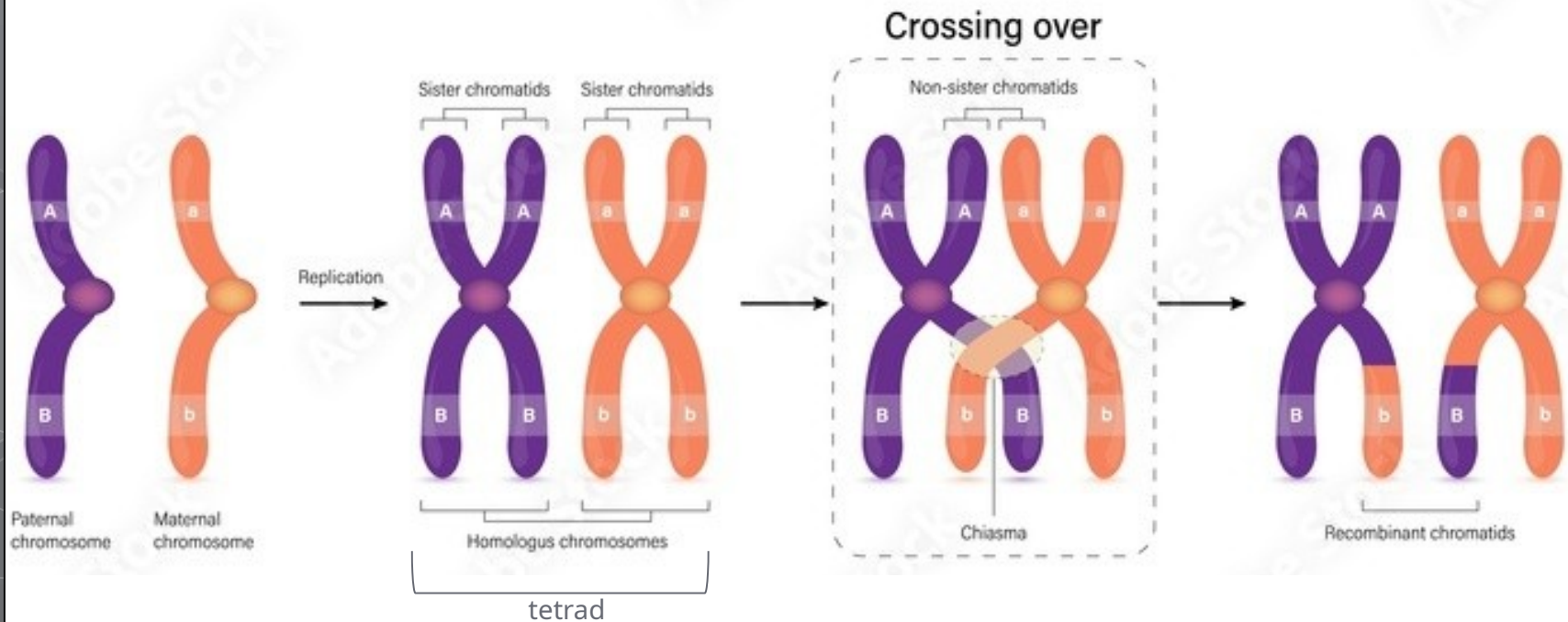
- **Chromatin condenses** to form chromosomes, spindle fibres form, **nuclear membrane dissolves**, etc.
- Homologous chromosomes pair up and align vertically (called ***synapsis***) and **crossing over** occurs



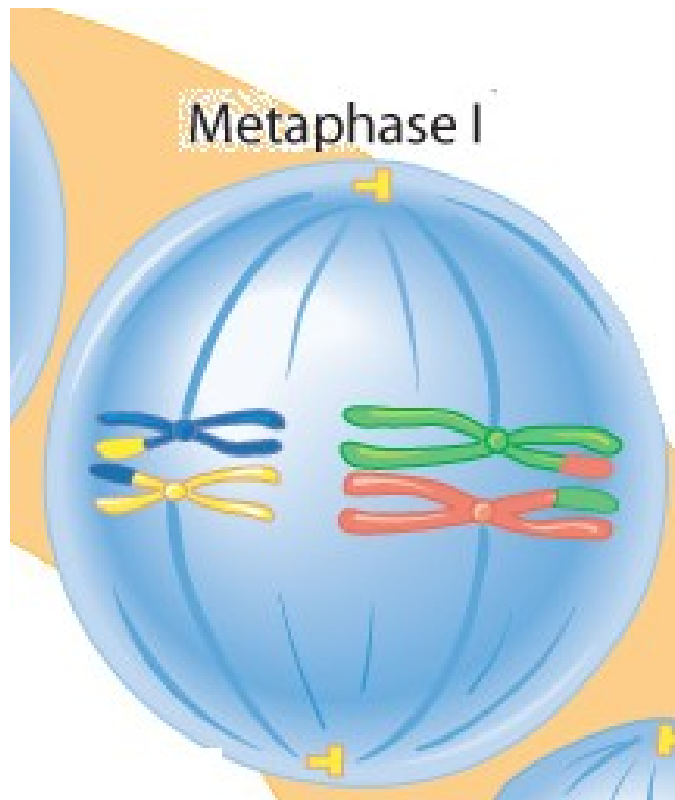
- **Crossing over** is an opportunity to mix genes.
 - This increases genetic variation and diversity which is important for sexually reproducing organisms
- Recall that homologous chromosomes are a set of the same chromosome, where one is maternal and one is paternal origin. These contain the same genes, but may be different alleles, or versions of those genes, such as the allele for 'blue eyes' vs 'brown eyes'



- Two homologous chromosomes aligned during synapsis is called a **tetrad**. Each tetrad contains 2 sets of 2 sister chromatids (total of 4 chromatids)
- Crossing over **occurs between non-sister chromatids** (there would be no point to cross over between sister chromatids as they are exact replicas of one another)



METAPHASE I



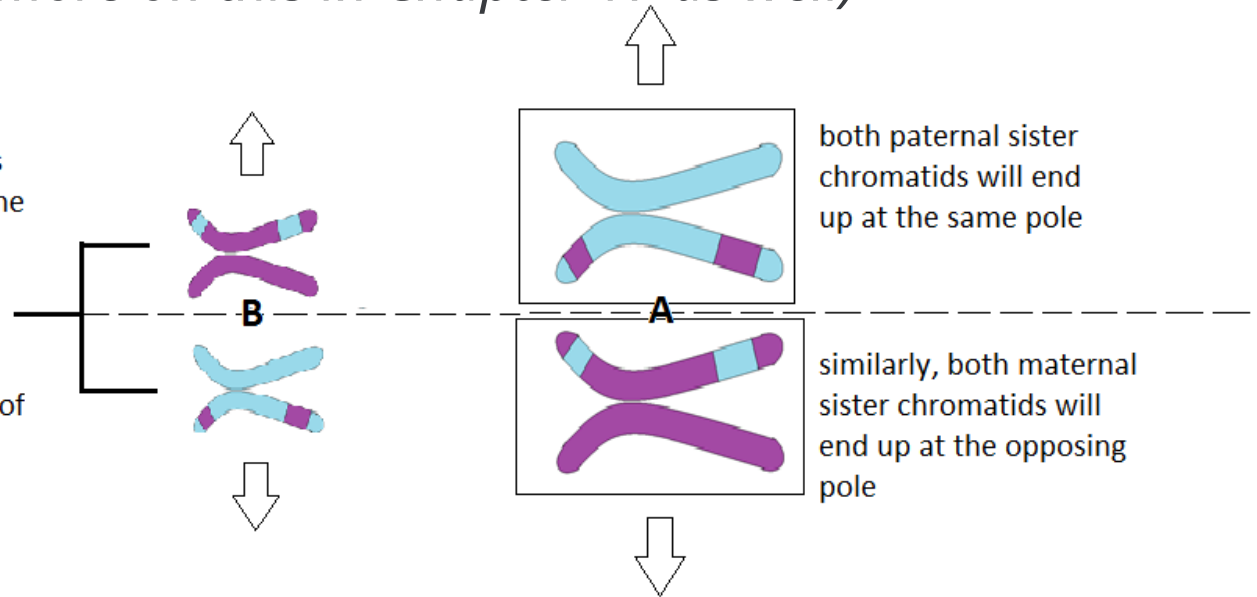
Key events:

- Spindle fibres attach to the centromere of each chromosome
- Instead of lining up in single file as they do in mitosis, **chromosomes line up as homologous pairs**, so that chromosomes of maternal and paternal origin will end up at opposing poles
- The orientation of each pair of homologous chromosomes is independent of the orientation of the other pairs (***independent assortment***)

Independent Assortment

(more on this in Chapter 17 as well)

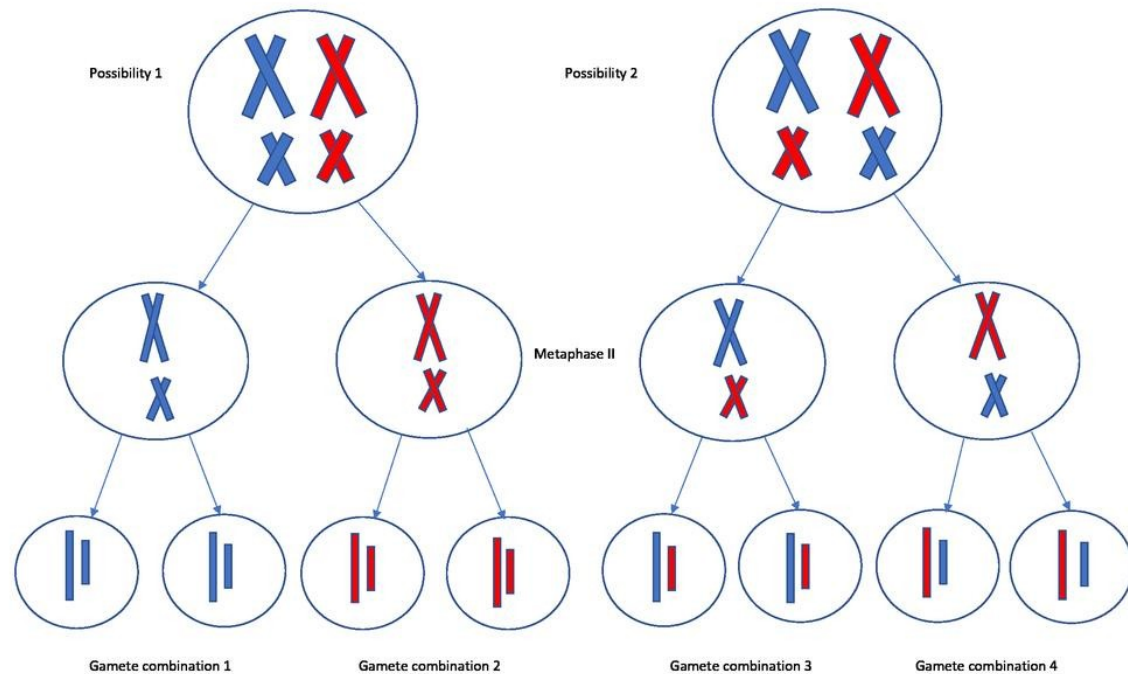
Different pairs of homologous chromosomes line up independently of one another (i.e just because paternal copies of chromosome A end up at the northern pole doesn't mean that paternal copies of chromosome B will end up there too)



Increases gamete diversity due to number of potential chromosome combinations

Independent Assortment

- The following diagram shows the possible gametes that can be formed from a $2n=4$ cell due to independent assortment.

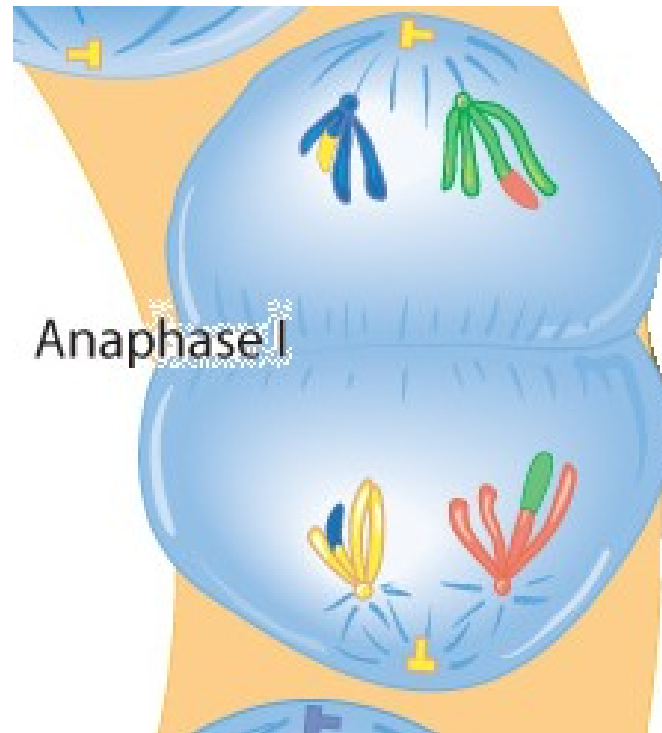


- Now imagine the number of possibilities for a human $2n=46$ cell!
 - To be precise, it would be $2^{23}=8388608$ possibilities! AND this is just one gamete (sperm or egg) that can join with a completely random gamete with the same 8388608 possibilities! That would be $8388608 \times 8388608 = 7036874418000$ ways to make a zygote between two parents. AND we haven't even considered crossing over!!

ANAPHASE I

Key events:

- Spindle fibres shorten, **pulling homologous chromosomes apart**
- Sister chromatids remain together, so that each new pole has a full set of DNA



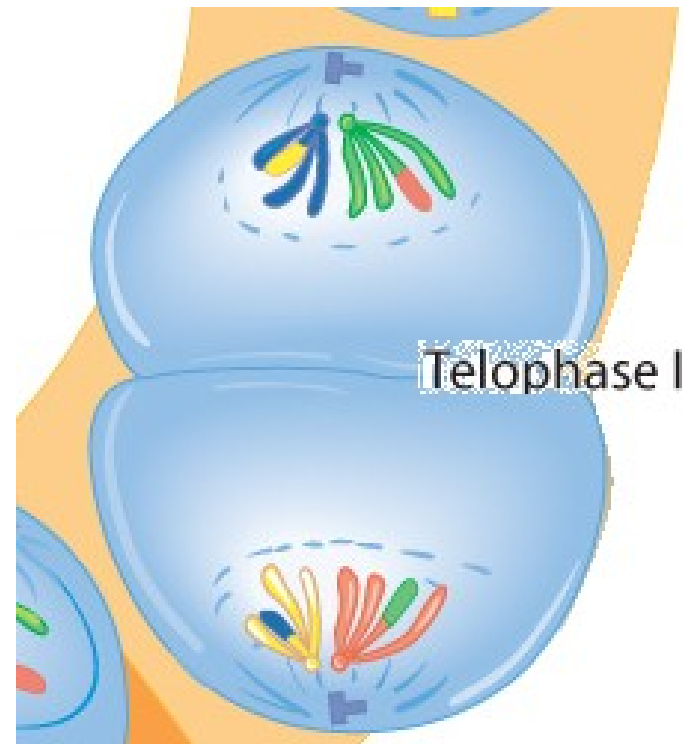
Important:

Anaphase I of Meiosis I is the **reduction step**. This means that the **daughter cells that are produced after Anaphase I are haploid**, meaning they only have one copy of each chromosome (i.e. $n=23$ in humans)

TELOPHASE I (and cytokinesis)

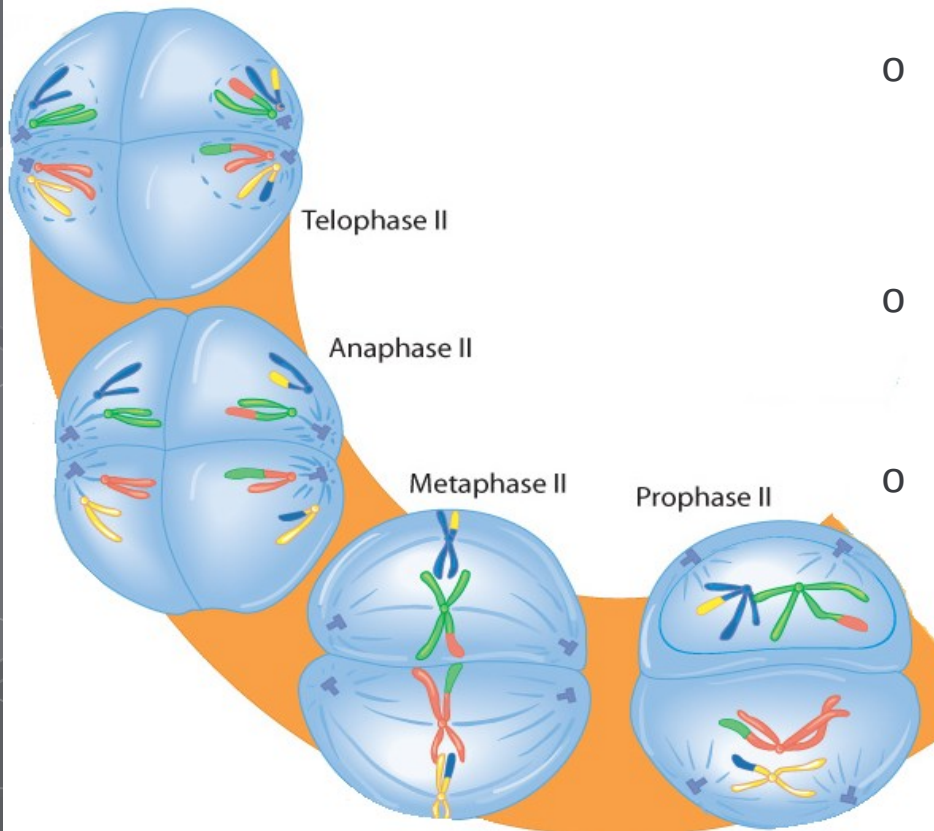
Key events:

- Homologous chromosomes begin to uncoil, spindle fibres disappear
- Cytoplasm is divided and a **nuclear membrane forms** around each new set of chromosomes
- Each new cell is **haploid** because they contain either the maternal or paternal set of homologous chromosomes (**reduction division** has occurred)



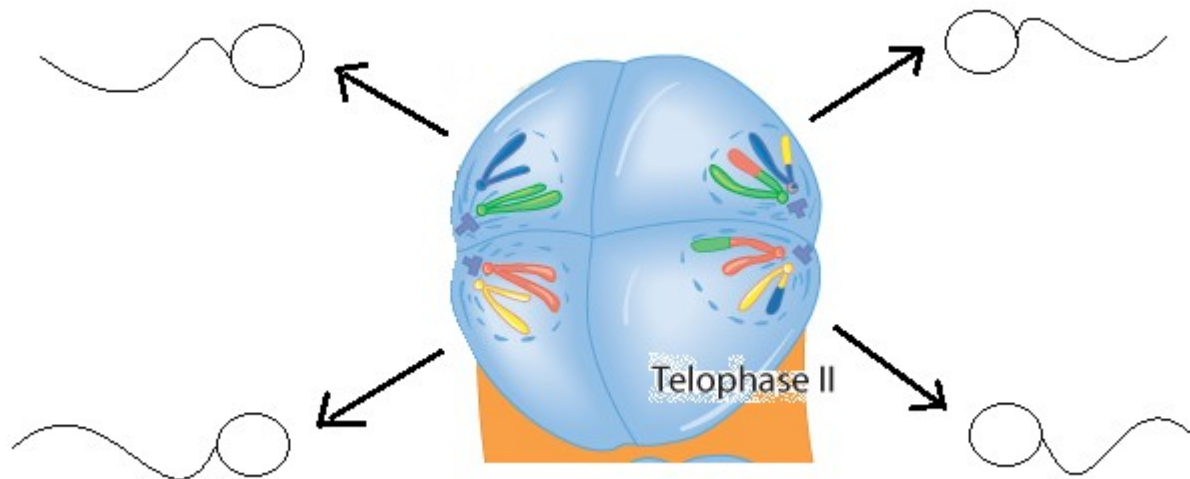
MEIOSIS II

Interphase does NOT occur before the cells enter prophase II (i.e. DNA is not replicated)



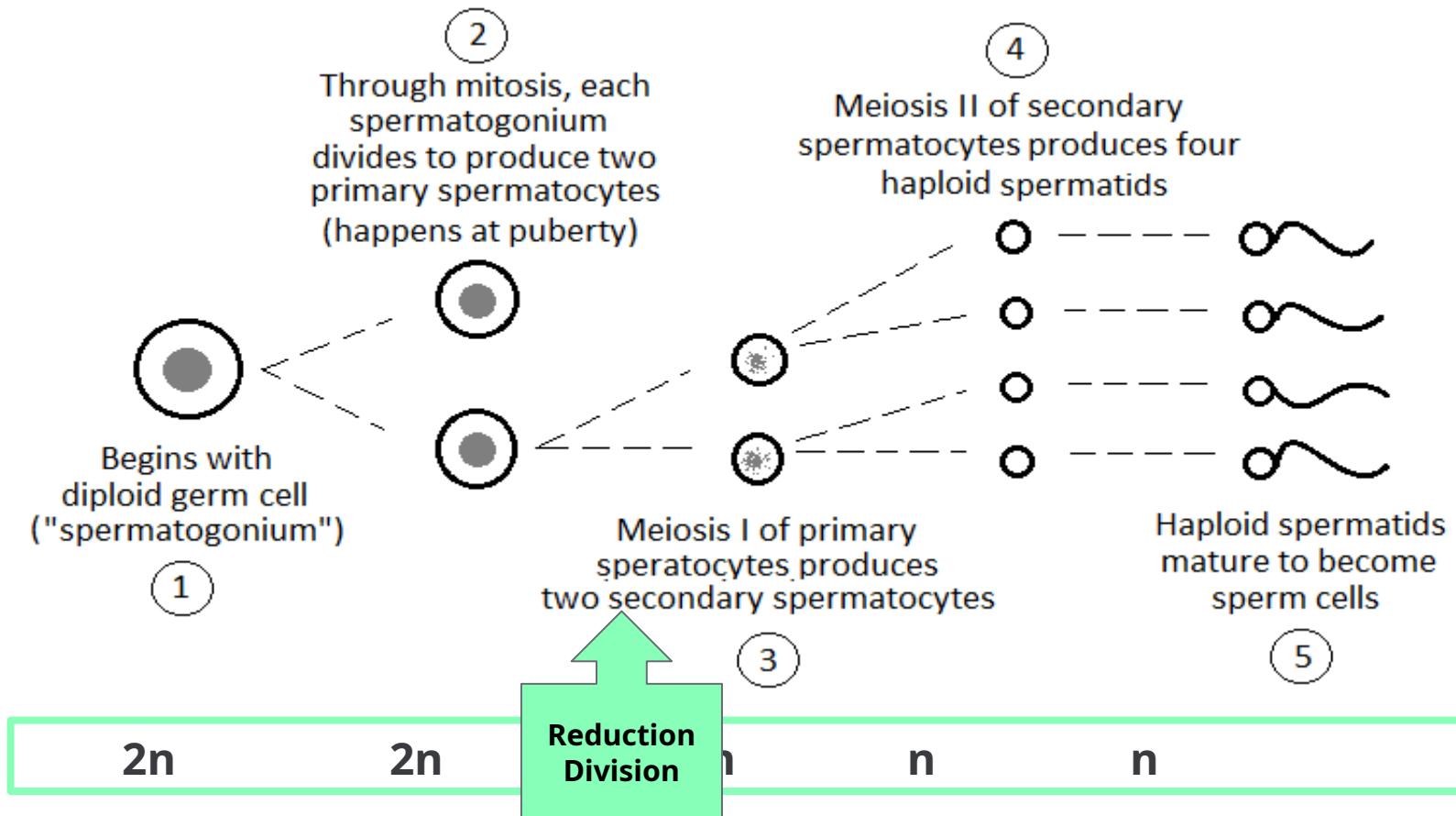
- Meiosis II is a lot like mitosis, with the exception of a haploid cell undergoing division to create identical daughter cells
- Cells proceed through prophase II, metaphase II, anaphase II, and telophase II
- The result is four haploid daughter cells with a single copy of each chromosome

Each daughter cell formed after telophase II will proceed through cytokinesis again to form the final gametes...

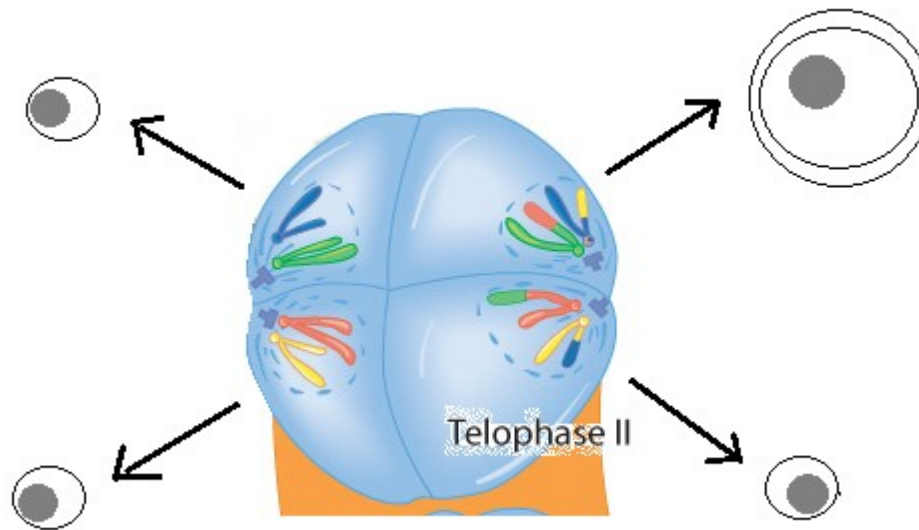


In males, this process is referred to as spermatogenesis, and results in the formation of four viable sperm cells

Spermatogenesis:

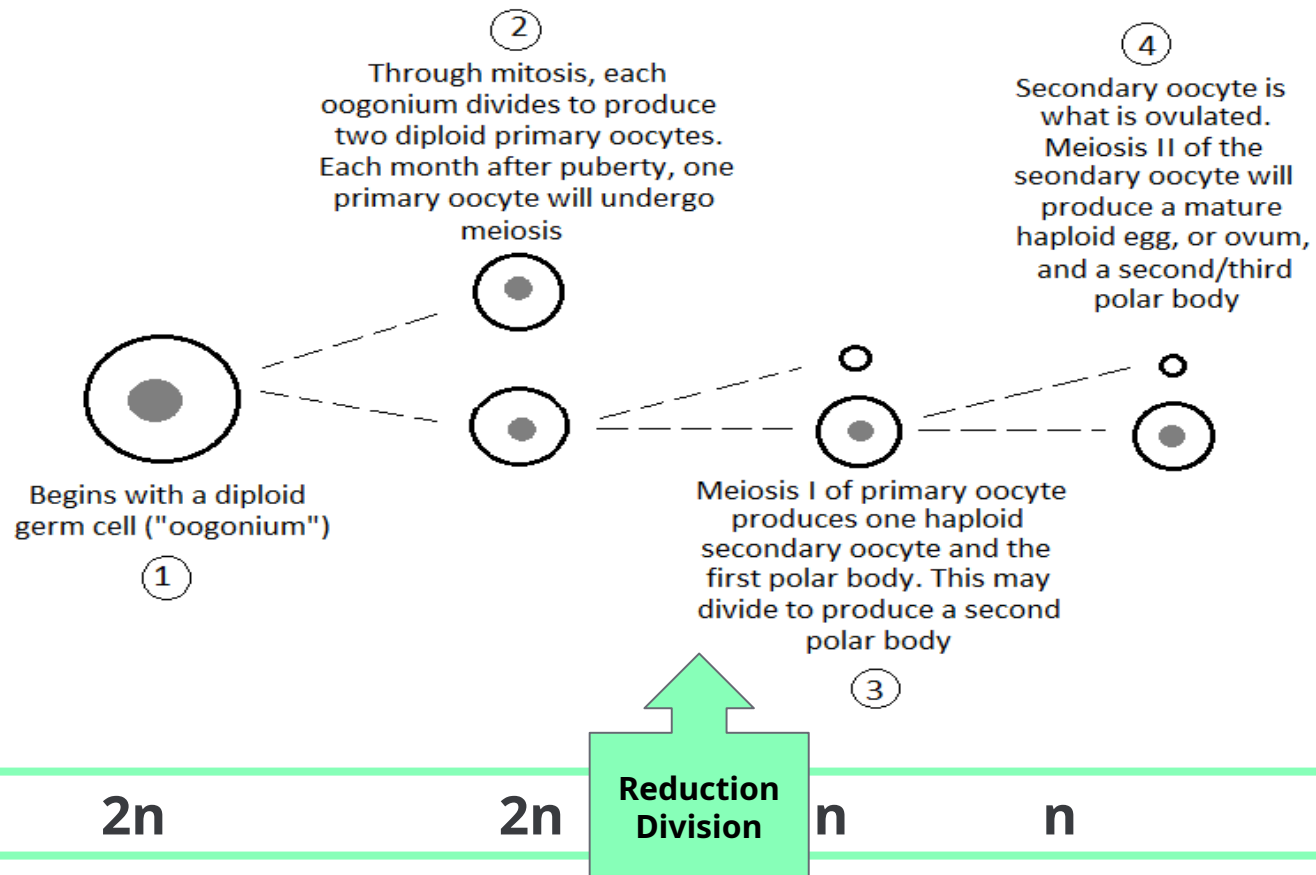


In females, the cytoplasm is not equally divided amongst gametes, resulting in a single viable ovum and three polar bodies:

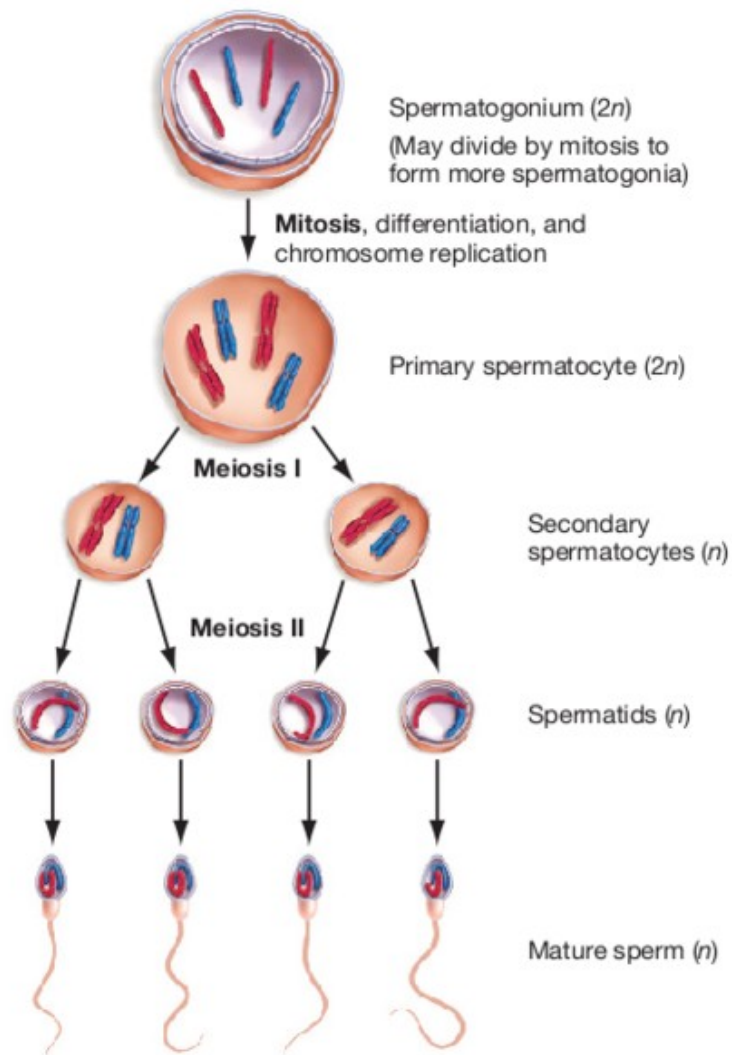


This process is referred to as oogenesis

Oogenesis:



(a) Spermatogenesis



(b) Oogenesis

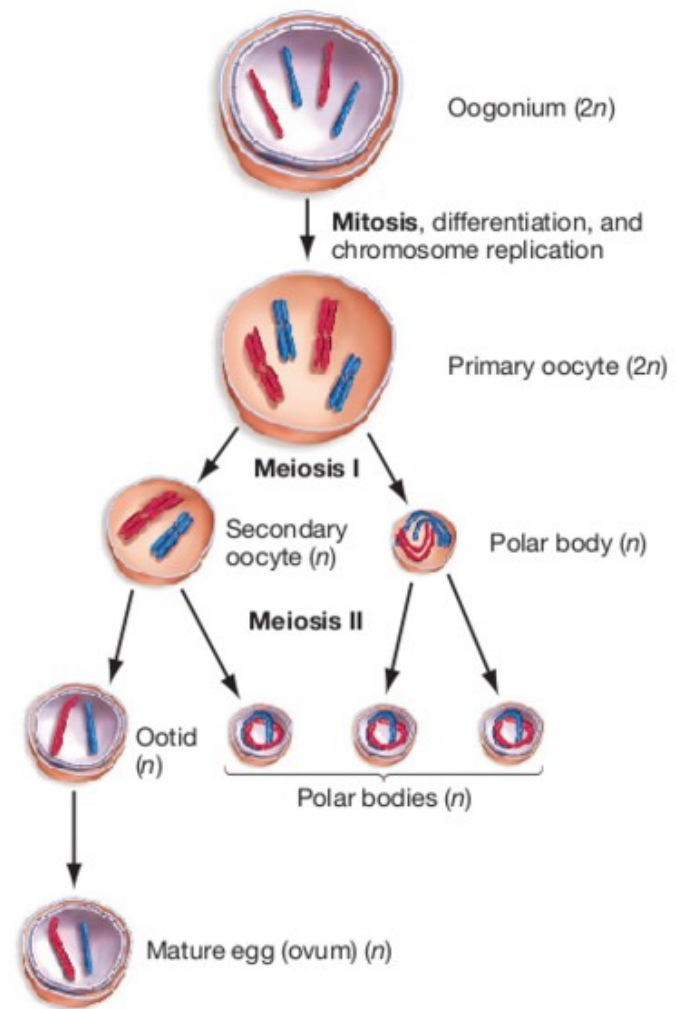
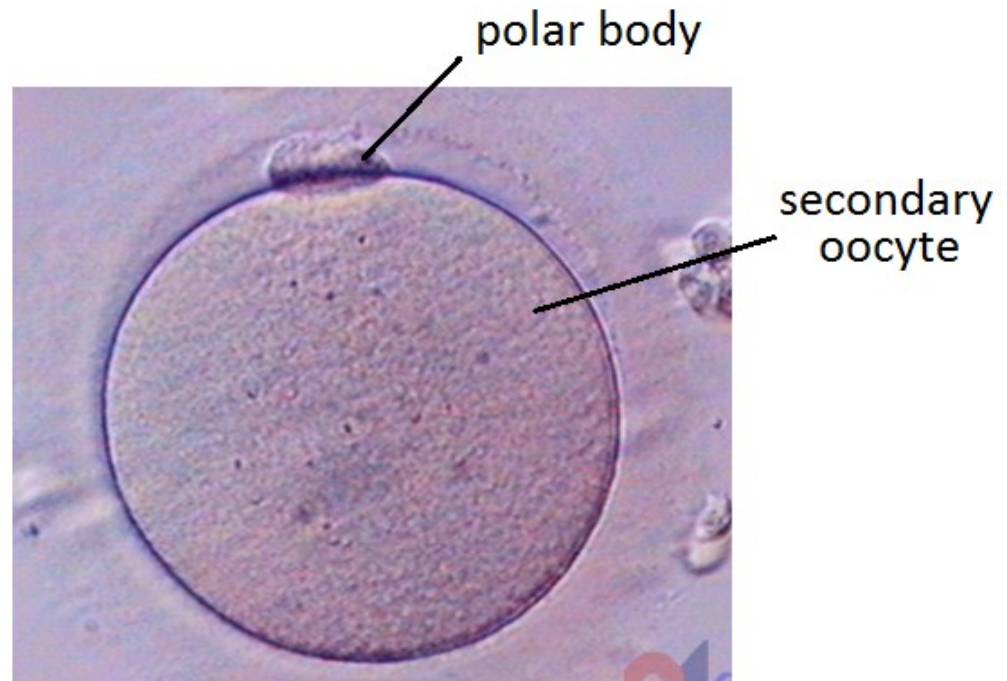


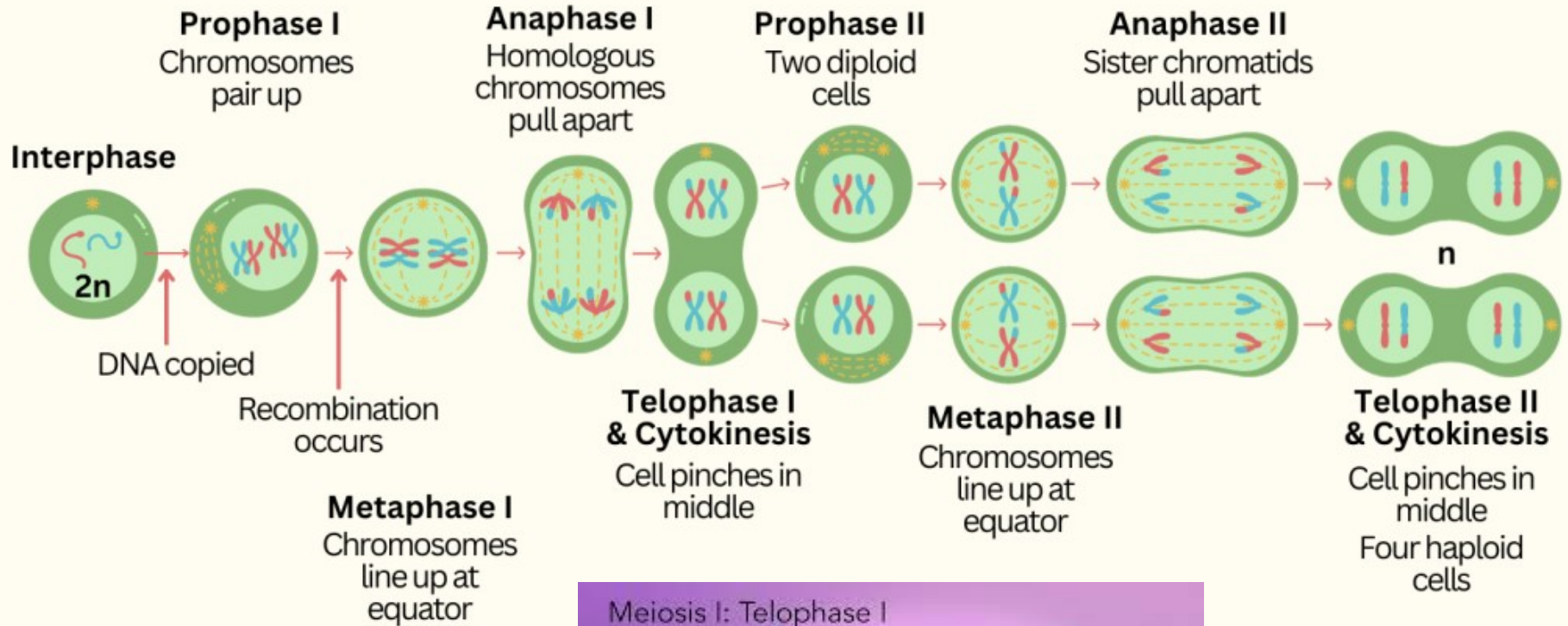
Figure 47.4 Gametogenesis in Sexually Reproducing Animals Produces Sperm or Eggs via Meiosis.

- Why does female gametogenesis involve unequal division of the cytoplasm? I.e. why are polar bodies made without any cytoplasm?

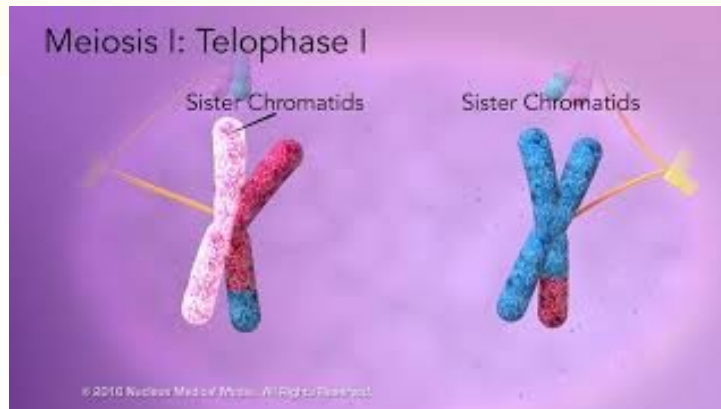


- This ensures that the single released ovum contains the maximum amount of cytoplasm including the organelles and nutrients. This will ensure that the developing morula and blastocyst has the means to survive until implantation occurs approx. 7 days after fertilization

Meiosis Recap

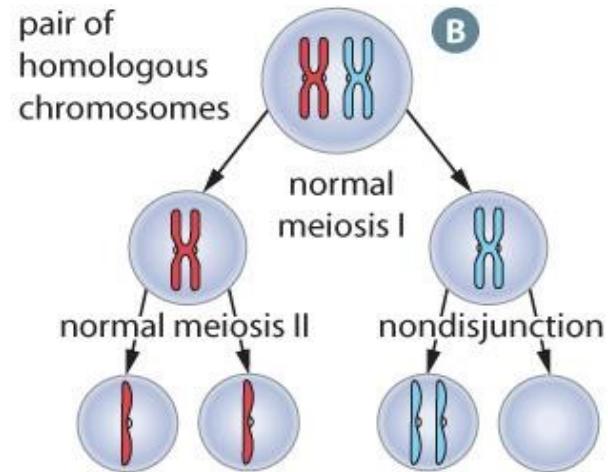
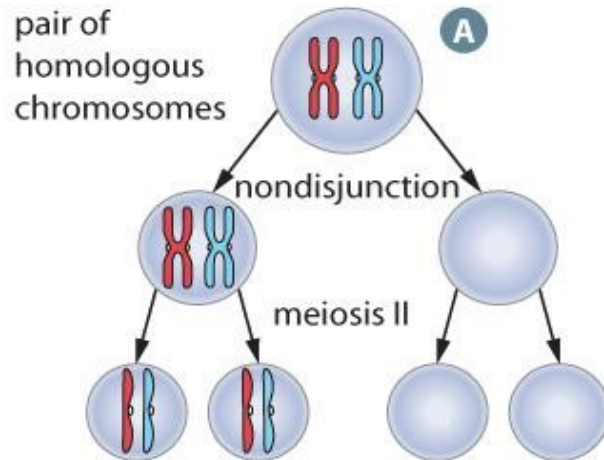


Meiosis I: Telophase I



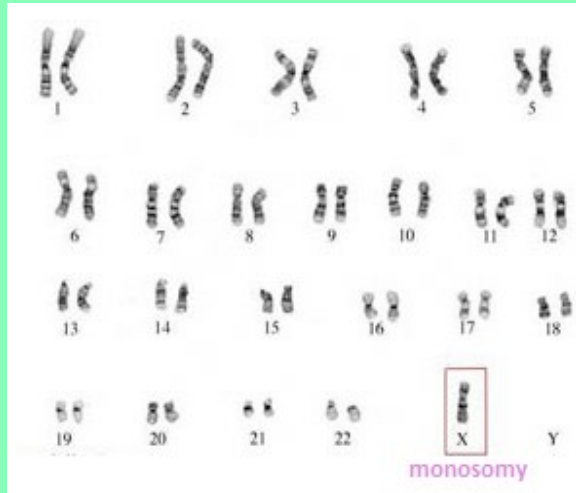
Nondisjunction: When things go wrong

You should be able to determine of disjunction occurred during meiosis I or II from similar diagrams



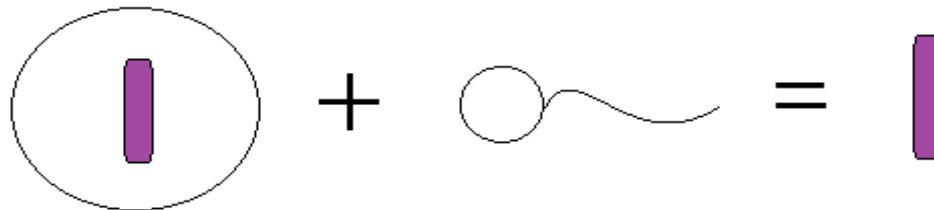
- Nondisjunction occurs when chromosomes do not separate properly during meiosis I or meiosis II
- Nondisjunction during meiosis I and meiosis II have different outcomes; however, both produce gametes with either too few or too many chromosomes:
 - **trisomy** results from having an extra chromosome while
 - **monosomy** results from having one less chromosome

Note: Trisomy and monosomy are not the same as being triploid or haploid. In triploid organisms there is an extra copy of every chromosome, whereas in trisomy there is an extra copy of one chromosome. Same situation for haploid vs. monosomy.

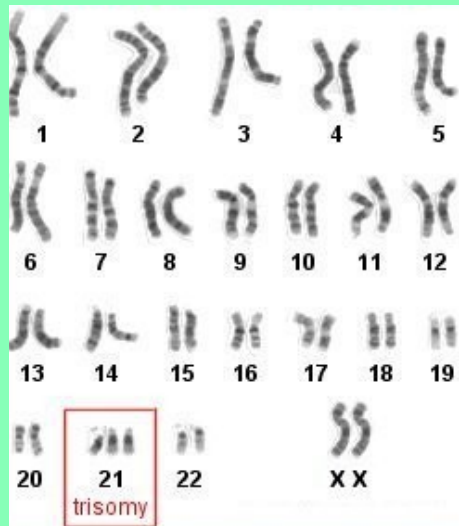


Example of a MONOSOMY:

Turner Syndrome (Monosomy X)

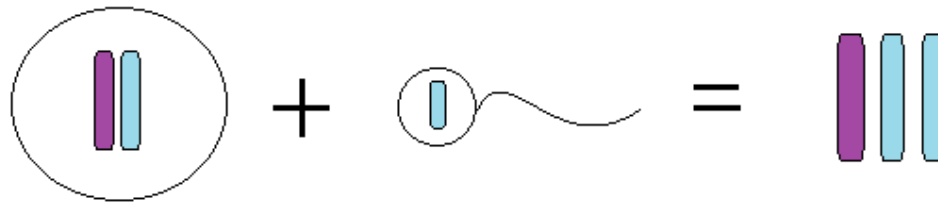


A normal, haploid egg is fertilized by a sperm lacking an x chromosome due to nondisjunction in meiosis II, resulting in a single x chromosome in each and every cell



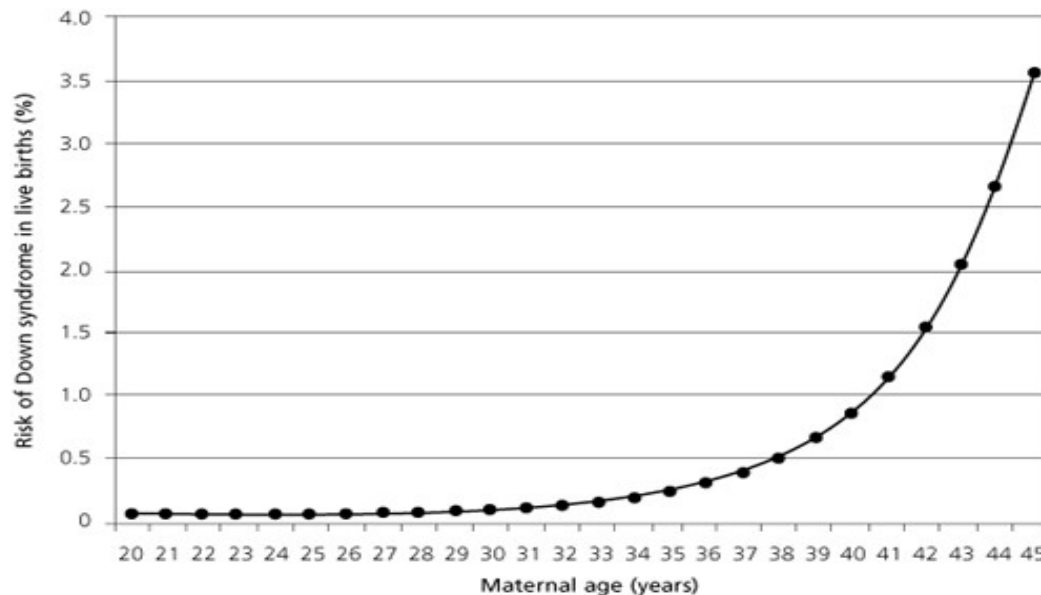
Example of a TRISOMY:

Down Syndrome (Trisomy 21)

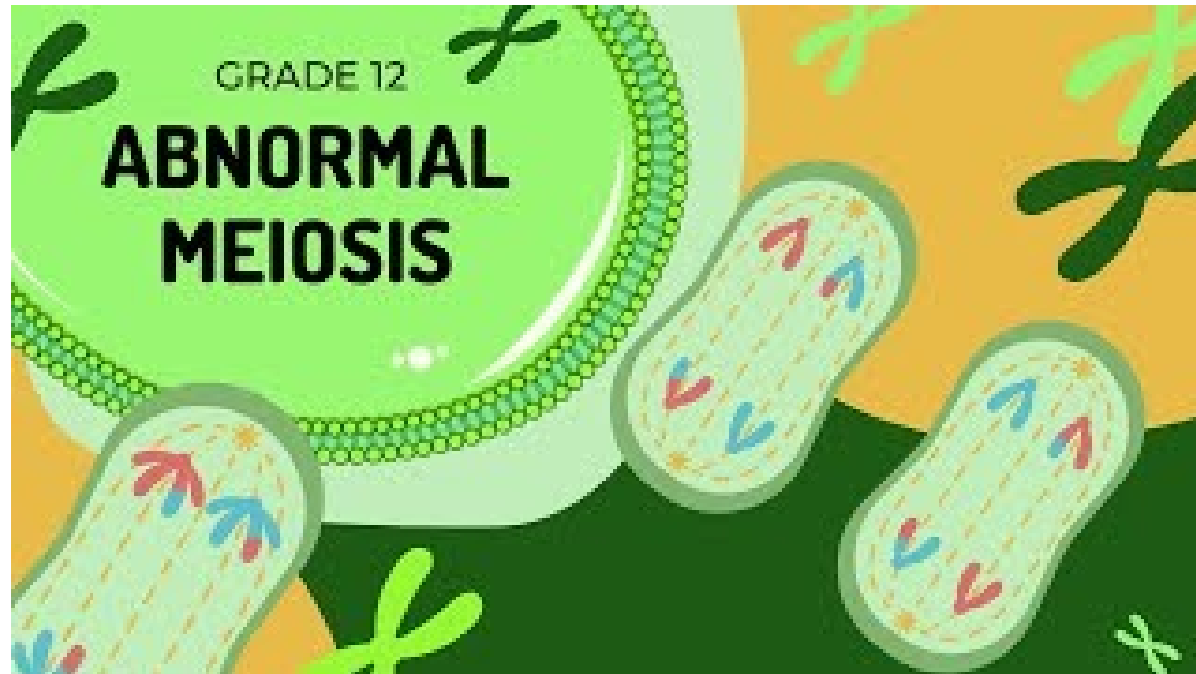


An egg with an extra chromosome 21 due to nondisjunction in meiosis I is fertilized by a normal, haploid sperm, resulting in an extra copy of chromosome 21 in each and every cell

Incidence of Down Syndrome with maternal age:



- As a woman ages, her probability of producing eggs with an extra copy of chromosome 21 increases dramatically. This is thought to happen because centromeres become “sticky” with age, meaning that nondisjunction will occur more readily
- Trisomy 21 can also be caused by nondisjunction in sperm cells



Applications of the Cell Cycle: Cloning

Cloning is the process of forming identical offspring from a single cell or tissue of a parent organ.

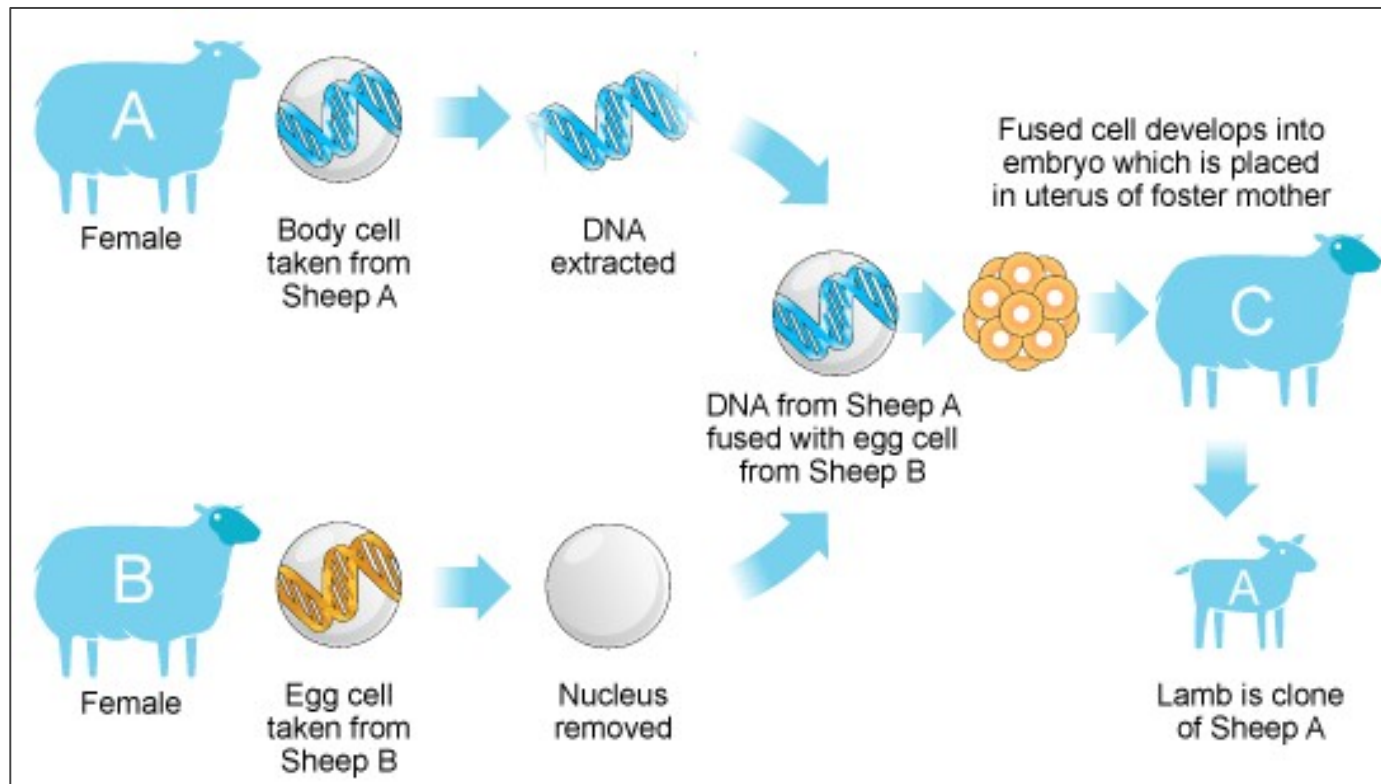
- **Cloning is a form of asexual reproduction.** Many species utilize this form of reproduction (e.g. plants, bacteria, hydra)
- A form of cloning also results in the production of twins; if a single fertilized egg accidentally undergoes mitosis and splits in two, **identical (maternal)** or **monozygotic** twins occur, each with the exact same DNA.
- **Non-identical (fraternal)** or **dizygotic** twins form when two eggs are ovulated at once, and are fertilized by different sperm. Dizygotic twins do not share the same genetic information.

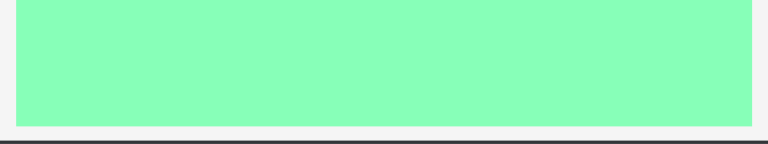


Animal Cloning Technology

(more on this in Chapter 18)

Organisms can be artificially cloned as well...





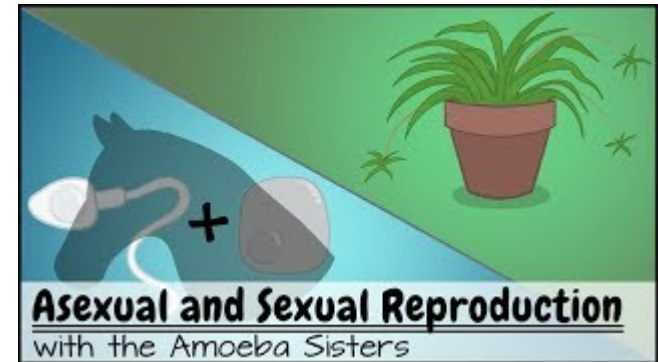
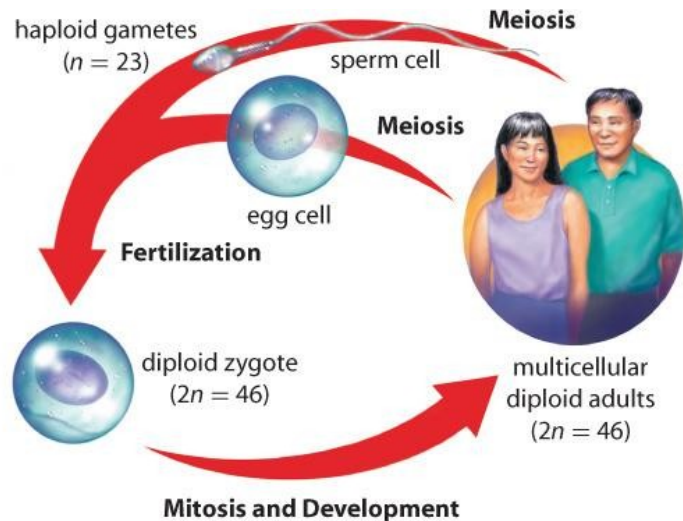
TOPIC 3:

Reproductive Strategies

- Reproductive strategies
- Alternation of generations

Textbook pgs. 573-583

Reproductive Strategies



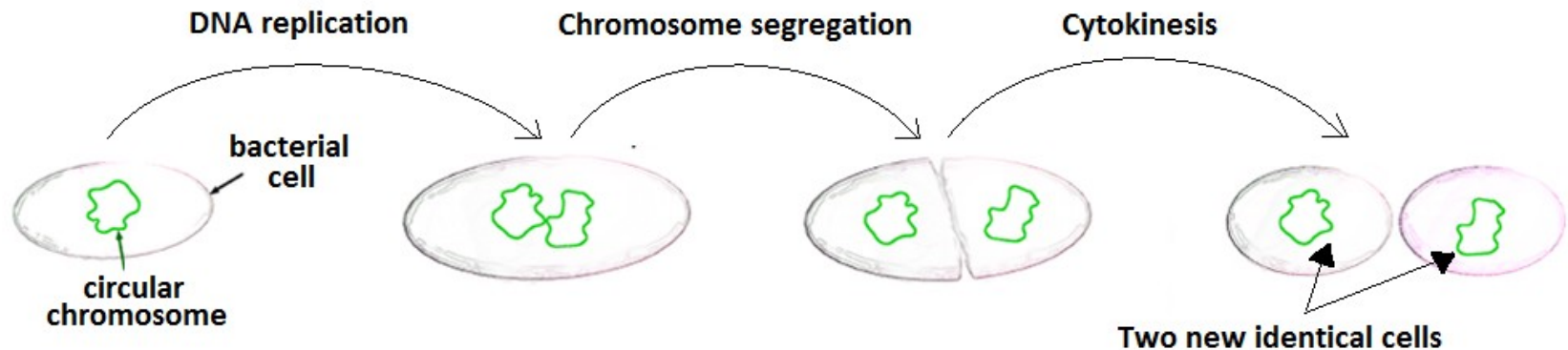
The human life cycle relies on regular patterns of meiosis and mitosis. However, other organisms rely on alternative reproductive strategies...

Asexual reproduction: Reproductive process where a parent organism produces genetically identical offspring, typically using mitosis or a similar process.

Sexual reproduction: Involves the production of gametes through meiosis, followed by fertilization between genetically distinct parental gametes to produce genetically distinct offspring.

Note: Over the rest of these slides you will be introduced to various reproductive strategies. You will not need to memorize and describe the various strategies. However, you will be expected to identify given a context whether it is sexual or asexual reproduction, where various events take place (mitosis/meiosis/fertilization) and the ploidy at each stage given context.

Binary Fission

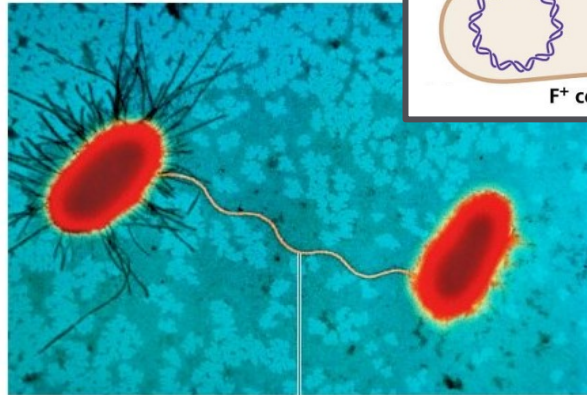


- Method of reproduction utilized by bacteria
- Allows for rapid growth of identical populations through asexual reproduction

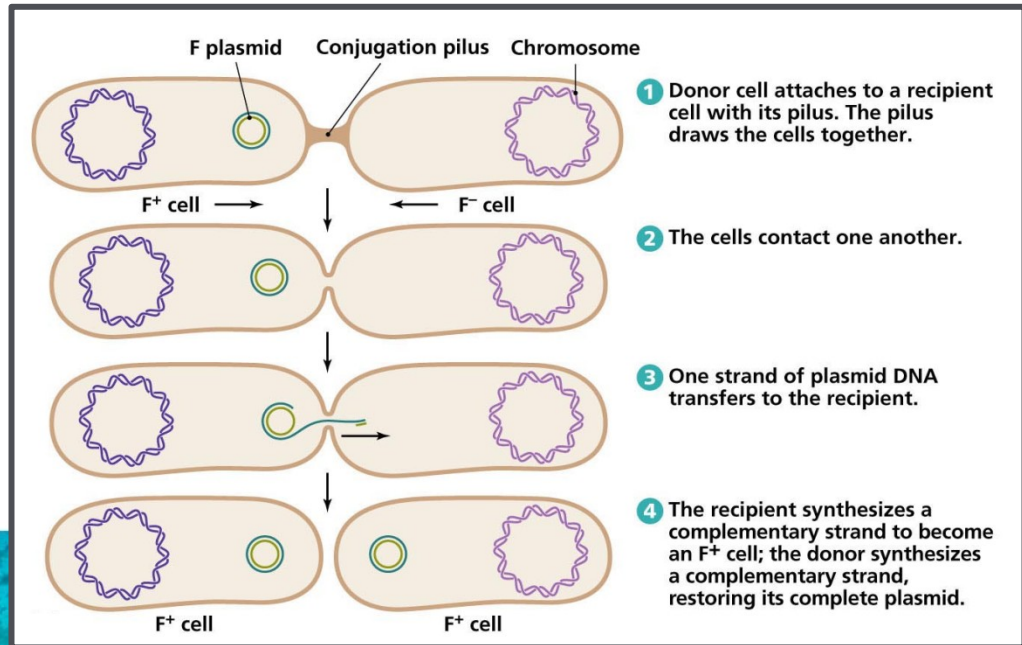
Advantages? Disadvantages?

Conjugation

- 0 Also utilized by bacteria
- 0 Plasmids (another source of DNA) are transferred from one cell to another through pili
- 0 Creates cells with new genetic combinations (kind of like sexual reproduction)



Sex pilus



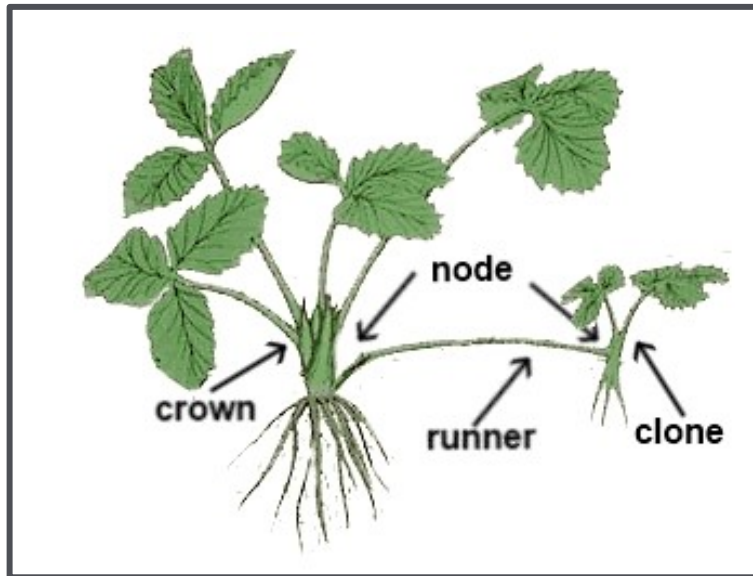
Budding

A form of asexual reproduction in which a complete but miniature version of the parent grows out from the parent's body; the newly formed clone separates only once it matures



Commonly utilized by metazoans such as corals, sponges, and hydra

Vegetative Reproduction



- o A form of budding that takes place in plants (asexual reproduction)



Parthenogenesis

A form of asexual reproduction in which an unfertilized egg develops into an adult



Example: Honeybees

- The queen bee lays both fertilized and unfertilized eggs
- Fertilized eggs develop into female worker bees, while unfertilized eggs develop into male drones

Spores

Form of asexual reproduction which allows parents to disperse their offspring over long distances

- o Often dispersed by wind or water
- o May be haploid or diploid
- o Some spores are actually the product of sexual reproduction (meiosis rather than mitosis), resulting in an alternation of generations



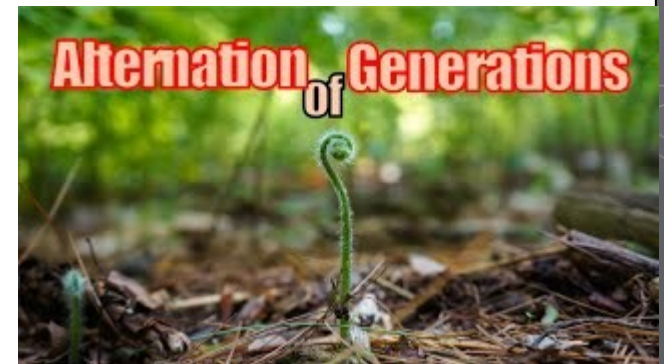
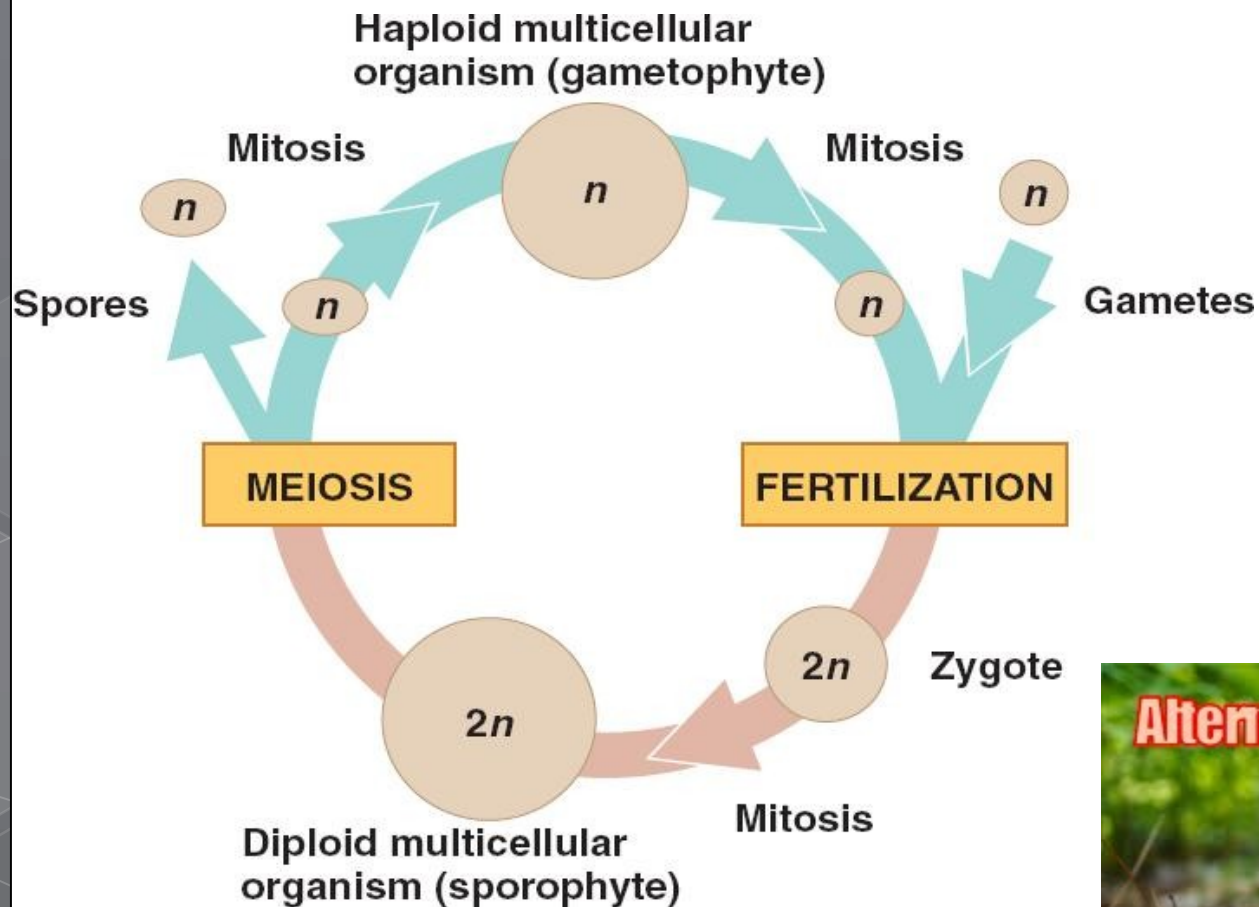
“ALTERNATION OF GENERATIONS”

The life cycle of some plants consists of two generations...

1. A diploid generation, or *sporophyte*
2. A haploid generation, or *gametophyte*

Through meiosis, the sporophyte (or spore-making body) produces multiple haploid spores, which spread and develop without fertilization. Each spore grows into a plant body called the gametophyte (or gamete-making body), which produces male and female gametes. Upon fertilization, gametes develop into another sporophyte, and the cycle repeats.

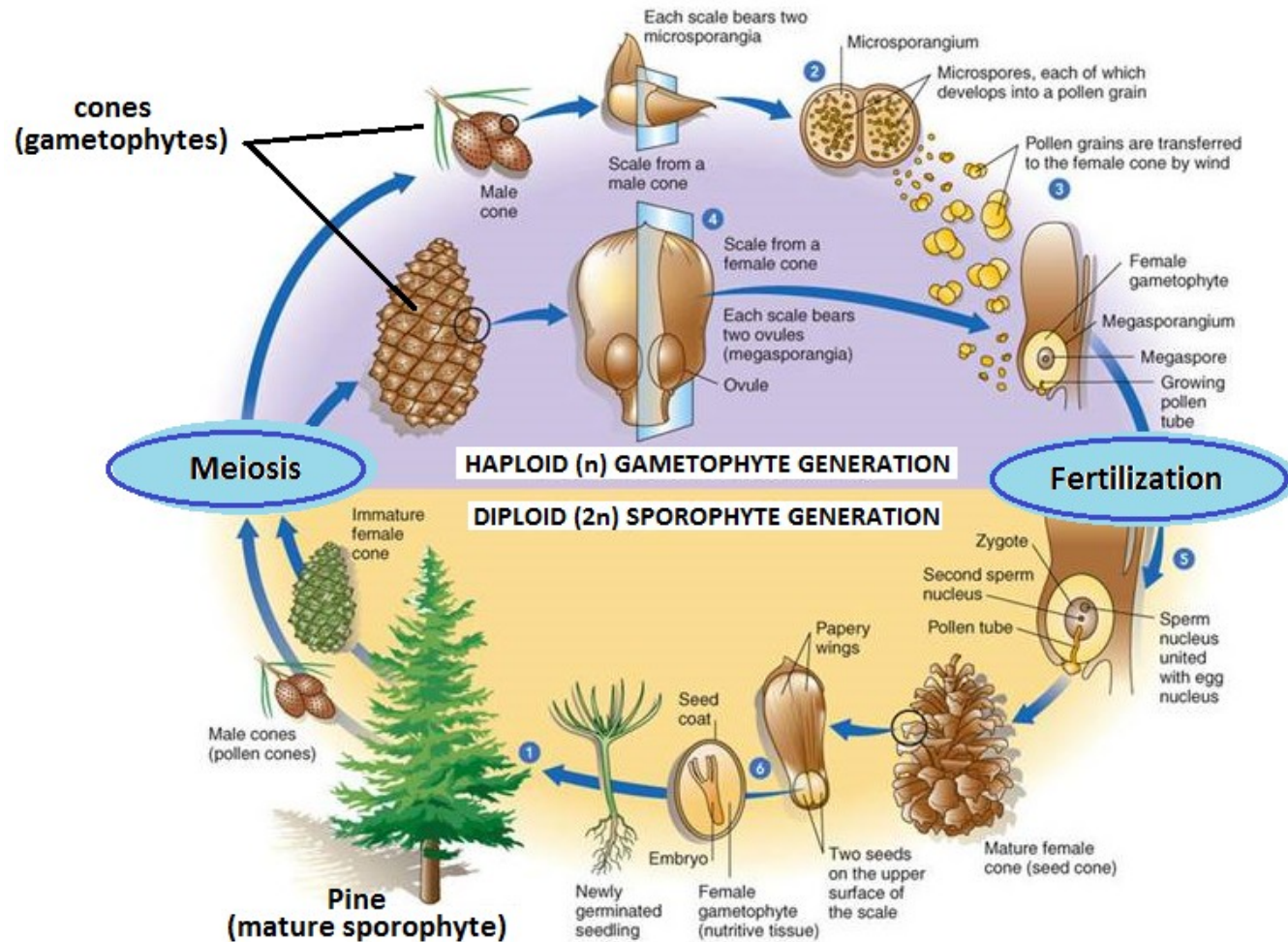
General model for alternation of generations:



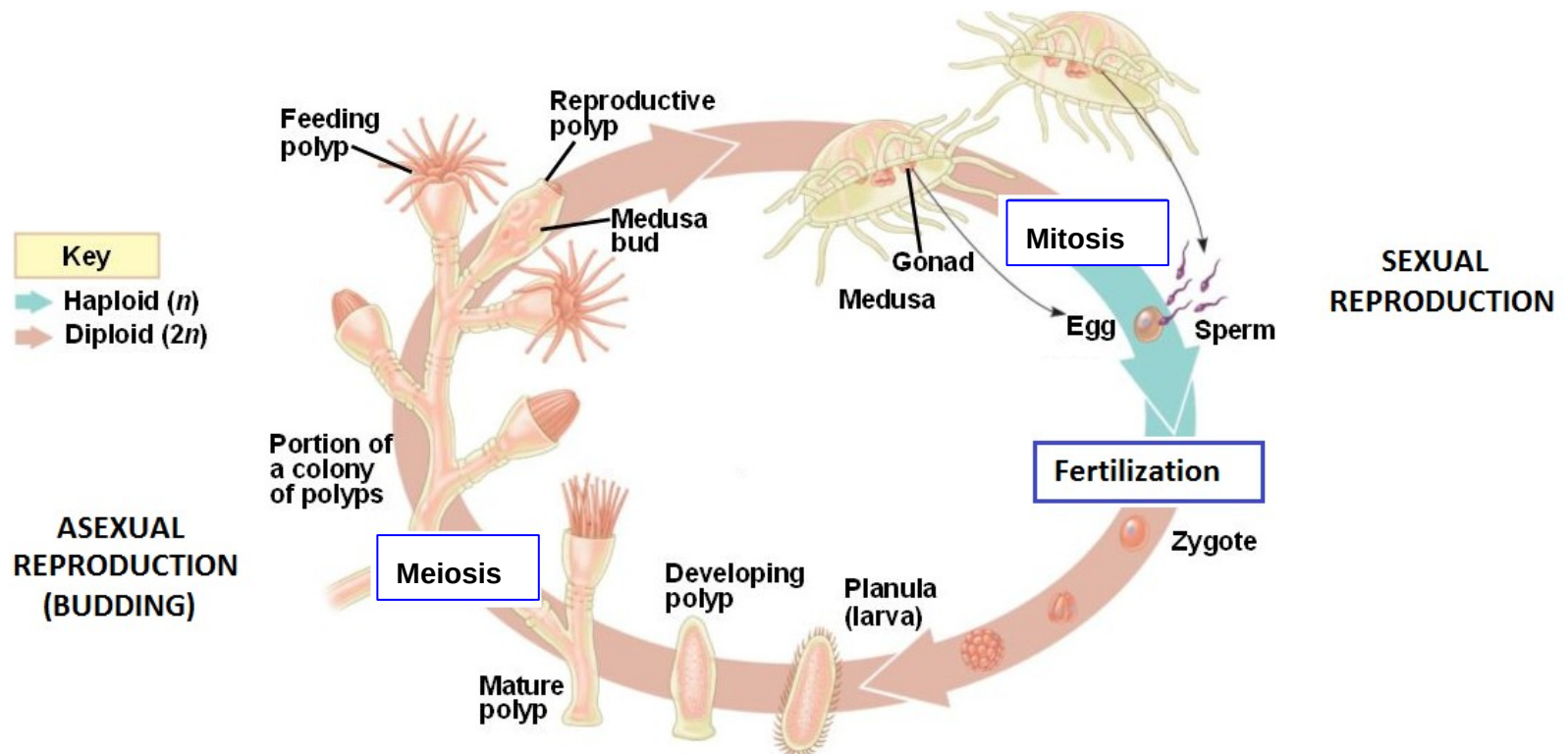
Many plants spend the majority of their lives in one phase or the other...

Conifers, for example, spend most of their lives in the diploid generation. The tree itself is this diploid sporophyte, while the cones contain the haploid gametophytes.



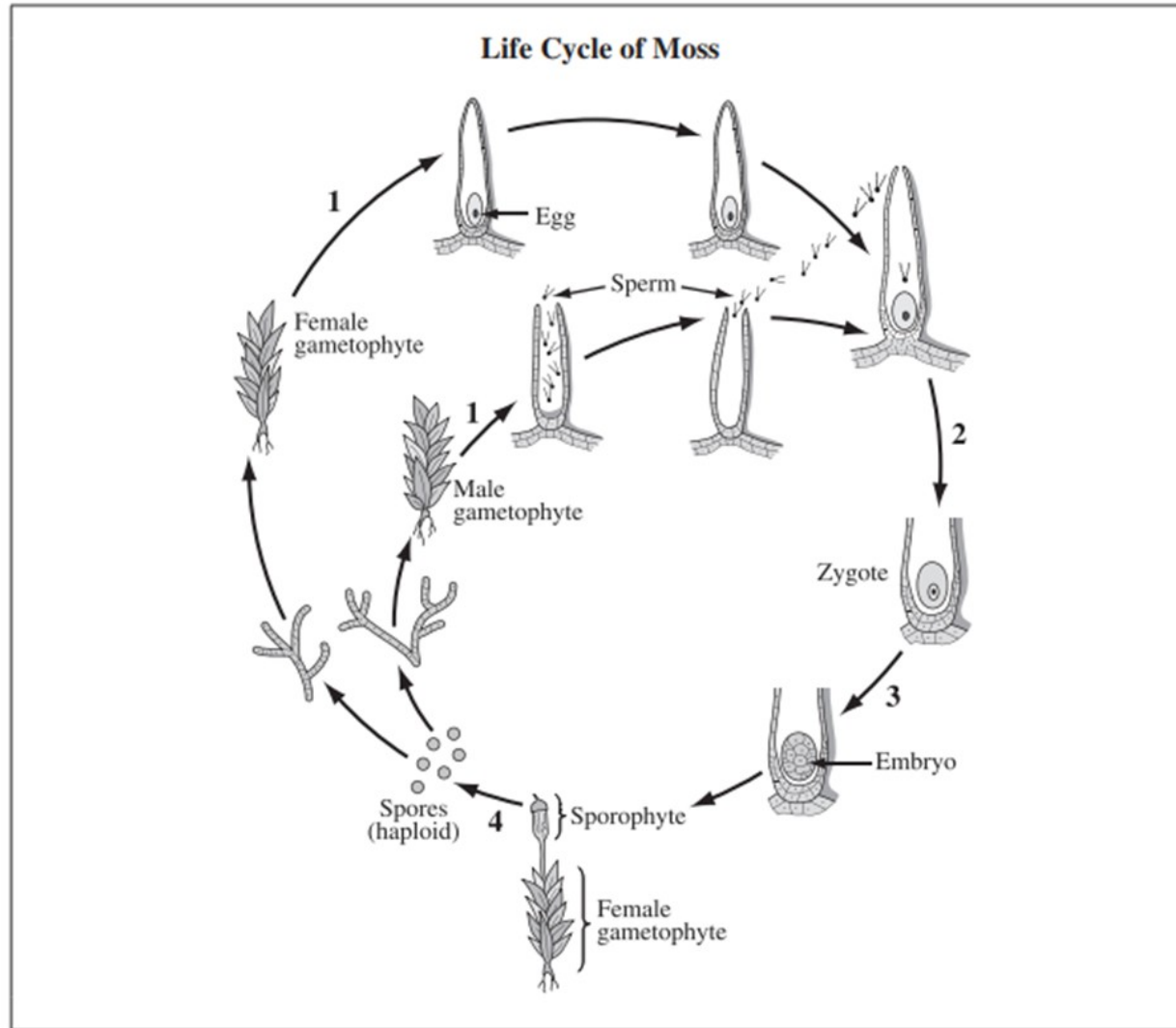


Rather than alternating between haploid and diploid generations, some organisms simply alternate between sexual and asexual reproduction (e.g. cnidarians):



Again, you do not need to memorize individual life cycles. Here are some example questions for what you may expect on an assessment:

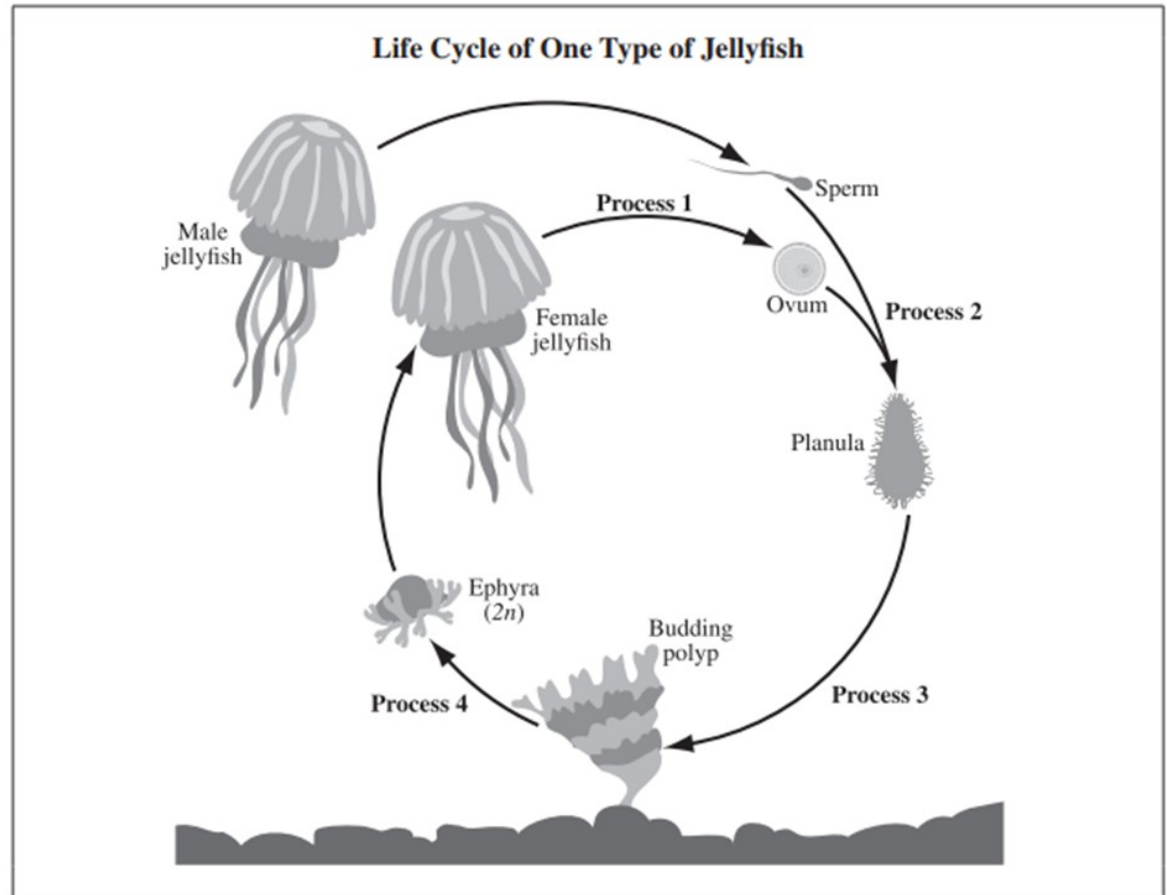
Process 2 is fertilization (sexual reproduction in this case) and process 4 is meiosis. The random combination of gametes and independent assortment of chromosomes during meiosis contribute to genetic variation.



68. Two processes in the life cycle of moss that contribute to increased genetic variation are

- A. 1 and 2
- B. 2 and 3
- C. 2 and 4
- D. 3 and 4

Process 2 must be fertilization as we observe gametes joining. This also indicates that the planula must be diploid ($2n$). Processes 3 and 4 must be mitosis as the ephyra is still diploid, therefore the budding polyp must also be diploid ($2n$). Finally, process one produces the gametes which must be haploid since when they join they create a diploid organism.



27. Which of the following rows identifies the ploidy of the planula, the budding polyp, and a gamete in the life cycle of a jellyfish?

Row	Planula	Budding Polyp	Gamete
A.	n	n	$2n$
B.	n	$2n$	n
C.	$2n$	$2n$	$2n$
D.	$2n$	$2n$	n